Electoral Cycles in Female Sterilization*

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Abstract

Many women across the world face substantial challenges in obtaining access to family planning — an important problem emphasized by the Sustainable Development Goals. The present study explores electoral cycles in female sterilization, the most prevalent contraceptive method in the world. We focus on Brazil, a country where many women have unmet demands for sterilization and often queue for many months or even years for their surgeries. Qualitative evidence suggests that local politicians distort the provision of publicly funded tubal ligation surgeries for political purposes. We analyze the universe of tubal ligation surgeries performed by the public health system, using regression analysis to examine variation across Brazilian municipalities in 1998-2019. We find that female sterilizations increase 8.8% during municipal election years. Moreover, they surge during electoral campaigns: female sterilizations increase 30.5% in the three months before municipal elections. Findings are similar when adjusting for overall hospitalizations, which rise less than 1% during municipal elections. Female sterilizations have more pronounced electoral cycles than do other elective surgeries (including vasectomies), and no cycles are detected for emergency surgeries. Electoral cycles in female sterilization are concentrated among nonwhite Brazilians, who face substantial health disparities. Female sterilizations increase far more during municipal than national elections, and they spike especially in poor municipalities and in the Northeast region. More broadly, our findings suggest that Brazilian politicians distort the public provision of contraception, an important issue not least because family planning is widely deemed a basic human right.

Keywords: Sterilization; Family Planning; Brazil; Electoral Cycles; Political Budget Cycles; Clientelism

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The freedom to choose the number and spacing of one's children is widely understood to be a basic human right.[1] For decades, international agencies have called for universal access to family planning: this objective was included in the 1994 International Conference on Population and Development, the Millennium Development Goals, and the Sustainable Development Goals. Despite marked improvements in recent years, many women across the world continue to face substantial challenges in obtaining access to family planning. Nearly a quarter of women globally who want to delay or avoid pregnancy have an unmet need for family planning, even though the number of modern contraceptive users has nearly doubled since 1990 (United Nations, 2022).

The present study focuses on female sterilization, one of the world's leading contraceptive methods. Despite recent declines, female sterilization continues to be the most prevalent form of contraception worldwide, used by 23% of married women of reproductive age, followed by condoms, IUDs and the pill — and is an order of magnitude more common than male sterilization (United Nations, 2022).[2] Unlike studies that investigate how politicians in various countries have coerced women to be sterilized,[3] we explore politicians' actions in a context where women have large unmet demands for sterilization. More specifically, we examine the case of Brazil, where women seeking tubal ligations from the public health system must often queue for many months or even years to be sterilized. This study is motivated by substantial qualitative evidence discussed below, which suggests that local politicians in Brazil often distort the provision of publicly funded tubal ligation surgeries for political purposes.

In this context with substantial unmet demand, our regression analyses reveal pronounced electoral cycles in female sterilization. We analyze the universe of tubal ligations provided by Brazil's public health system across over 5,500 municipalities between 1998 and 2019. Leveraging both across-year and within-year variation, we estimate whether tubal ligation rates rise in election periods, controlling for long-term trends, seasonality, and municipal characteristics. We show that over these two decades, the public provision of tubal ligations

increased by 8.8% during municipal election years. Even more remarkably, this increase was concentrated in the three months immediately preceding municipal elections, when female sterilizations surged by 30.5%. Results are similar when controlling for broader electoral cycles in inpatient procedures, which increase by less than 1% during municipal elections. While the present study focuses on female sterilization, we also provide novel evidence of significant but considerably smaller electoral cycles in vasectomies. We likewise find significant but proportionally smaller electoral cycles when examining other elective surgeries with long queues, but no such cycles when examining emergency surgeries. Electoral cycles in female sterilization are concentrated among nonwhite Brazilians, who face substantial health disparities. Female sterilizations increase far more during municipal than national elections, and they surge especially sharply in poor municipalities and in the Northeast region.

These findings contribute to the broader literature on political budget cycles. Many scholars, building on seminal works of Nordhaus (1975), Tufte (1975), and Rogoff & Sibert (1988), have investigated the logic and conditions that lead politicians to shift fiscal and other policies according to the timing of elections. Numerous empirical studies suggest that incumbents increase expenditures just before elections, at both the national and subnational level.[4] Politicians often increase their provision of targetable benefits that are highly valued by voters (Brender & Drazen, 2005; Drazen & Eslava, 2010). Our quantitative findings add to this literature by showing that political budget cycles similarly affect the public provision of female sterilizations in Brazil, where the procedure is not only a leading contraceptive method but is also one of the most demanded elective surgeries. While our results are consistent with theories that suggest politicians distort expenditures during election campaigns to signal competence or demonstrate policy preferences to voters (e.g., Persson & Tabellini, 2000; Rogoff, 1990), they are also consistent with Hanusch & Keefer's (2014) argument that clientelism often contributes to political budget cycles. In line with that possibility, qualitative evidence indicates that some Brazilian politicians offer sterilizations in *quid pro quo* exchange for votes during election campaigns. As discussed below, while our broader findings

suggest that Brazilian politicians distort the public provision of female sterilization, further research is needed to parse the mechanisms underlying how and why they do so. The Conclusion emphasizes that the welfare effects of these electoral cycles deserve further examination, as they may involve the net expansion of sterilizations, as well as their reallocation across time and/or patients.

1 Context and Qualitative Evidence

Our study focuses on electoral cycles in tubal ligation surgeries performed by Brazil's public health system (commonly known as SUS for *Sistema Único de Saúde*, or Unified Health System). The 1988 Constitution guarantees universal, comprehensive healthcare, and most Brazilians — especially those who are poor or nonwhite — obtain medical care exclusively from SUS.[5] According to the nation's census bureau, nearly 72% of Brazilians rely exclusively on SUS for their healthcare; among black and brown Brazilians, this share is 80% and 81%, respectively (IBGE, 2020).[6] Given our focus on how sterilizations surge during municipal elections — held concurrently every four years nationwide to elect mayors and city councilors[7] — it deserves emphasis that municipalities have considerable resources and discretion with respect to SUS expenditures. Each level of government has duties in administering SUS, with transfers to municipalities accounting for 34% of public health care expenditures in 2022.[8] Furthermore, federal audit court reports suggest that municipal officials have significant autonomy when providing some healthcare services, frequently with little oversight by states (TCU, 2011).

Over the past three decades, female sterilization in Brazil has transformed from an illegal (albeit prevalent) method to a publicly funded method of contraception. Tubal ligation — a surgical procedure that aims to prevent pregnancy by cutting, sealing or tying a woman's fallopian tubes — was illegal and banned by the medical ethics code until 1997 without "precise indication approved by two medical doctors." Though illegal, sterilizations were not criminalized, and doctors commonly employed successive cesareans as a rationale for

performing tubal ligations (Caetano & Potter, 2004). In a radical departure from previous policy, and after substantial effort from civil society organizations, Law 9263 legalized and provided public funding for sterilization. This law specified that recipients must be at least 25 years old or already have two living children, and must undergo a 60-day waiting period with required consultations. After the passing of Law 9263, doctors performed 1,060,549 publicly funded tubal ligations between 1998 and 2019. These procedures are performed in a hospital setting under anesthesia and typically require inpatient care, trained surgical staff, and access to an operating room. During our study period, the number of female sterilizations performed by SUS increased for a decade before stabilizing.[9] As shown in Figure 1 and explored more rigorously below, this provision of tubal ligations increases substantially during municipal election campaigns.

[Figure 1 about here]

Despite expanded access in recent years, there is substantial unmet demand for female sterilization. Across the public healthcare system, SUS users often complain about accessibility and long waiting times (SIPS, 2011). In a representative survey conducted by the Federal Council of Medicine (CFM) in 2018, 68% of respondents said it was "difficult" or "very difficult" to obtain surgeries from SUS (CFM, 2018).[10] Queues were the most common complaint, with 61% reporting that wait times for surgeries were "bad" or "terrible." Moreover, 39% of respondents reported that they were currently waiting for an appointment or surgery from SUS: of these, 29% had already waited a year, and another 16% had waited 6-12 months. Administrative data similarly show long queues. An unprecedented CFM study discovered there were waiting lists for 904,000 elective surgeries in 2017, based on data reported from states representing only half of Brazil's surgeries (CFM, 2017a, 2017b).[11] Of these patients, 59% had already queued for six months, and 13% had queued for 18 months. Data from Minas Gerais provide a glimpse into the scale of unmet demand for tubal ligations: in 2017, the state had 11,249 women on the waiting list for the procedure — over double the number actually performed in Minas Gerais that entire year (4,905).[12] Numerous studies

and journalists have documented the long waiting times facing women who seek sterilizations from the public health system.[13] More broadly, Brazil's government recognizes this major issue and has taken strides to reduce waiting times for elective surgeries, including the launch of a "National Queue Reduction Program" in 2023.[14]

As this study examines rigorously, amidst this substantial unmet demand and lengthy queues, politicians distort the timing of when public hospitals perform female sterilizations for electoral purposes. Brazil's electoral laws forbid politicians from engaging in such practices, regardless of whether tubal ligations are provided to signal competence, to demonstrate policy preferences, or in *quid pro quo* exchange for votes (i.e., clientelism). During election years, Law 9504 prohibits politicians from distributing free goods and services beyond what was already budgeted during the prior fiscal year, except during certain emergencies. Moreover, Brazil's 1965 Electoral Code prohibits clientelism: it is a violation of criminal law "to give, offer, promise, solicit or receive, for oneself or for another, money, gifts, or any other benefits, in order to obtain or give a vote, or to obtain or promote abstention, even if the offer is not accepted." Beyond criminal charges, electoral judges can impose swift administrative sanctions through Law 9840 (Reis, 2006; Tozzi, 2008): over a thousand politicians were ousted between 2000-11 for electoral clientelism, the top reason for politician removals in Brazil (Nichter, 2021).

While the quantitative analyses below lack data to investigate whether electoral cycles involve clientelism, qualitative evidence suggests that the clientelist use of female sterilization has long precedent in Brazil.[15] In 1993, a national Parliamentary Inquiry Commission (CPI) focused on tubal ligations more broadly concluded that "the use of surgical sterilization in exchange for votes is now a frequent occurrence in various parts of the country" (Congresso Nacional, 1993). The CPI, which urged criminal investigations, based its assessment on evidence uncovered by state-level commissions, numerous congressional witnesses, and press reports. Journalists have also long reported on the issue. As one example, a 1990 news article entitled "Clientelism: Candidate Ties Tubes in Pirambu" — one of Brazil's

largest favelas — led to criminal investigations against a doctor-candidate who reportedly sterilized women "in exchange for votes" and affixed campaign paraphernalia on three cars transporting "electoral clients on the day of surgery."[16] Caetano & Potter (2004), who study the period before female sterilizations were legalized in Brazil, also provide evidence of their clientelist provision during election campaigns. This practice continued after Law 9263 legalized and provided public funding for female sterilization. Indeed, a leading television news program (*Fantástico*) reported on the use of female sterilization for vote buying in 2012,[17] and prominent anti-clientelism organizations even publicly warned voters of the practice in 2010: "If you are ever offered sterilization or tubal ligation in exchange for your vote ... report it!"[18] As explained by an ex-federal deputy in 2011, many politicians "have highly vulnerable voters, who depend on clientelist actions ... In general, it's an individualized good: I give you a tubal ligation, a job, and you give me a vote."[19]

To glean further insight, consider several examples of Brazilian politicians convicted over the past two decades for the clientelist provision of female sterilizations. According to Brazil's highest court, a mayoral candidate engaged in this practice during the 2004 campaign in Marabá, Pará.[20] The candidate's campaign recruited 13 women (with the help of his stepdaughter, a candidate for city council) to receive free tubal ligations in direct exchange for their votes. They were sterilized by the candidate's son-in-law and another doctor in a local hospital. Contrary to the legally mandated 60-day waiting period, some of the women testified that they underwent surgery the same day they signed up. Two women testified that they regretted being sterilized and had not received the required consultations to review alternative contraception methods. A judge declared that the candidate's "electoral corruption ... surpassed the limits imaginable of humanity," sentencing him to house arrest for three years and a monetary fine.

In Porto Velho, Rondônia, a city councilor candidate was sentenced to three years in prison for buying votes with female sterilization in the 2004 electoral campaign. According to the judicial ruling, the candidate explained to numerous women that he was "scheduling

tubal ligation surgeries free of charge for patients, so long as they showed support for said candidate, voting for him in that year's elections and getting family and friends to do the same."[21] Women who agreed to the terms had to provide their voter registration numbers and other personal details, and were told when to arrive at his campaign office for transportation to the hospital. Numerous women testified that they exchanged their votes for sterilization, and that the candidate — who was a doctor and former city councilor — performed the surgeries himself.

In Goianápolis, Goiás, a city councilor candidate was convicted of clientelism during the 2008 campaign. According to court documents, the candidate promised a free tubal ligation surgery to a woman in exchange for the votes of her and her husband.[22] When driving her to the hospital a few weeks before the election, the candidate emphasized that he needed their votes, and discussed how he frequently helped others to become sterilized. A complication during the tubal ligation surgery led to the discovery of this vote-buying scheme; the municipality's Secretary of Health became suspicious because the surgery had not been arranged following protocol. Upon questioning, the woman admitted the *quid pro quo* nature of the exchange. When issuing its final verdict, Goiás's state electoral court ruled that "there was no other motive for the tubal ligation surgery than to acquire the vote and support of the voter and her family." The candidate was removed from office, fined, and became ineligible to run for elected office for eight years.

In Rio de Janeiro state, a politician running for state deputy in 2010 employed tubal ligations for clientelist purposes, just a few years after being ousted as mayor for vote buying.[23] During the 2010 election, the candidate and his brother — both physicians at a medical clinic associated with the public health system — were each convicted of providing tubal ligations to women in exchange for their votes. Witnesses reported to police that while at the clinic, they were instructed to vote for the candidate in exchange for their sterilizations, received his campaign materials, and were asked for their voter documents. A 29-year-old who received a tubal ligation reported being told that she would get "the operation for free,

if I would help him elect his brother."[24] Upon conviction, electoral authorities overturned the politician's election and declared him ineligible to run for public office for eight years.[25]

As a final example involving clientelism, a city councilor in São José de Ribamar, Maranhão was found guilty of the clientelist provision of sterilizations during the 2020 election. According to court documents, he spearheaded "an illicit scheme to schedule appointments and surgeries in the public health system ... in exchange for the promise of votes."[26] Search warrants revealed that his campaign possessed medical consultation forms that listed not only citizens' names but also detailed voter registration information. Moreover, authorities discovered audio recordings in which staff reminded tubal ligation recipients not to "forget" their "commitment to us," and also discussed the need to "demand votes from the people on the list" so they would "honor the word they gave to vote for the candidate." Maranhão's state electoral court removed the city councilor from office, imposed a fine, and ruled him ineligible to hold public office for eight years.

Taken together, the qualitative evidence presented thus far suggests that some Brazilian politicians engage in the clientelist provision of tubal ligations during elections. However, clientelism is by no means the only explanation for electoral cycles in female sterilization. As discussed in the Introduction, an extensive literature on political budget cycles — which does not focus on clientelism — investigates other key reasons why politicians often have incentives to increase expenditures during campaigns. Some quantitative studies, which focus on neither clientelism nor healthcare, identify political budget cycles in Brazil (e.g., Klein and Sakurai, 2015; Sakurai and Menezes-Filho, 2011). Moreover, when we interviewed 71 local politicians and other elites in Northeast Brazil, many noted that incumbents temporarily provide more healthcare and other services before elections.[27] For example, a city councilor explained: "Generally, the mayor, during election time, does many programs for the community — assistance for everyone who needs healthcare."[28] Likewise, a party leader explained that "healthcare, when it's election time, always has better service. Sometimes they place more doctors, they treat people better, for fear of losing votes."[29] To the ex-

tent that incumbents undertake such actions, electoral cycles may be observed even without clientelism.

With respect to female sterilization in particular, qualitative evidence also discusses electoral cycles without specifically mentioning *quid pro quo* exchanges. In a leading newspaper in 1986, a doctor involved in a public health commission explained that tubal ligations were "always" used "during election periods ... to obtain votes of the poorest population," and a nurse claimed that "female sterilization substantially grows during electoral campaigns."[30] In a study by Caetano (2000), one sterilized woman explained that: "it was election time and during these periods many doctors and politicians do this and that for free. Tubal ligations, especially, were at the peak! The day I went to have my surgery there were about 12 women who would have tubal ligations."

Altogether, this qualitative evidence sheds light on politicians' use of female sterilization for political purposes — only some of which involves clientelism (i.e., direct, contingent exchange). Consistent with both clientelist and non-clientelist interpretations, quantitative analyses below reveal electoral cycles in female sterilizations. We emphasize at the outset that our regressions do not enable us to parse the mechanisms underlying why the public provision of tubal ligation surgeries surges during election campaigns. As discussed in the Conclusion, further research is needed to examine the extent to which sterilizations increase due to clientelism versus other reasons for boosting expenditures during elections such as generating goodwill, conveying policy effectiveness, and demonstrating policy preferences.

2 Data

To investigate the dynamics of public healthcare provision, we employ rich administrative data from DATASUS, collected by Brazil's Ministry of Health. We use Hospital Admission Authorizations (*Autorizações de Internação Hospitalar*, or AIHs), which are mandatory forms that healthcare providers must submit when admitting patients in order to receive reimbursement from SUS. These data provide detailed case-level information on the universe of

hospitalizations in Brazil's public healthcare system, including location, timing, procedures performed, and demographic characteristics of admitted patients.[31] We use the SUS data to construct municipal-level procedure rates, with our key dependent variable measured as the number of tubal ligation surgeries per 100,000 women of reproductive age (i.e., 15 to 49 years), employing 2000 population data from Brazil's census bureau (*Instituto Brasileiro de Geografia e Estatística*, or IBGE). We also show robustness by employing the municipal-level count of tubal ligations as the dependent variable. While the tubal ligation rate is our primary outcome of interest, we also compute procedure rates (using the medically relevant population as the denominator) for several other common inpatient procedures, involving both elective surgeries (vasectomies, hernia repairs, gallbladder removals, and cataracts) and emergency surgeries (appendectomies and heart attack treatments) — as well as the overall number of inpatient procedures (AIHs). We examine hospitalizations between January 1998 and December 2019, spanning the period during which Brazil's government broadly provided public female sterilizations and before the COVID-19 pandemic. Hence, our analysis examines a monthly panel of Brazilian municipalities covering 22 years.[32]

Our analysis is centered around municipal elections, as local politicians and officials play a pivotal role in administering public healthcare services in Brazil. According to the Ministry of Health, municipalities are responsible for implementing the public health system in their territory, using their own resources as well as federal and state transfers, and they play a role in formulating healthcare policies.[33] Local politicians thus have considerable influence over the timing and volume of healthcare provision. Our primary specifications focus on tubal ligations performed on women residing in any of Brazil's 3,006 municipalities with a hospital.[34] This sample reflects 78% of publicly provided sterilizations in 1998-2019 and comprises the tubal ligations most subject to influence by local politicians.[35] Furthermore, we focus our analysis on tubal ligations performed separately from a cesarean delivery, as the timing of such procedures is anchored to the timing of childbirth and is therefore less susceptible to electoral cycles. However, we underscore that all findings hold when investi-

gating the universe of all publicly provided tubal ligation surgeries. As shown in Appendix Tables A3, A7, A9, and A11, results are similar and remain significant when analyzing all female sterilizations across all of the more than 5,500 municipalities in Brazil.

3 Methodology

3.1 Electoral Cycles in Female Sterilization: Annual Variation

We employ two approaches to show the robustness of our results. The first empirical strategy estimates the relationship between the timing of elections and rates of tubal ligation using year-level variation. This approach simply assesses whether election years are associated with higher levels of female sterilizations, controlling for broad time trends and municipality characteristics. More specifically, we estimate the following regression:

$$y_{mt} = \tau \text{ Election Year}_t + \beta X_{mt} + \theta_1 \text{Year}_t + \theta_2 \text{Year}_t^2 + \gamma_m + \varepsilon_{mt} \tag{1}$$

where y_{mt} is the monthly tubal ligation rate in municipality m and month t (per 100,000 female residents of reproductive age). Election Year$_t$ is an indicator equal to 1 in years when municipal elections are held (i.e., in 2000 and every subsequent four years). The vector X_{mt} includes time-varying covariates such as municipal GDP per capita and the total hospitalization rate (AIH rate). Municipality fixed effects γ_m control for time-invariant characteristics of municipalities, and a quadratic time trend accounts for secular national trends in procedure rates. Ancillary analyses also include an indicator equal to 1 in years when national elections are held (i.e., in 1998 and every subsequent four years).

3.2 Electoral Cycles in Female Sterilization: Within-Year Variation

Our second empirical strategy focuses on identifying within-year cycles in the provision of tubal ligations around municipal elections. Rather than comparing average rates across election and nonelection years, we exploit the fixed timing of elections in Brazil to examine

how procedure rates vary across the calendar year — specifically, whether sterilizations spike in the months leading up to an election.

We leverage a convenient feature of the Brazilian electoral system to estimate the effects of electoral cycles. In Brazil, municipal elections are held on the first Sunday of October, every four years.[36] This institutional regularity allows us to credibly model counterfactual trends, using data from nonelection years to account for secular time trends and the seasonality of tubal ligations. We examine deviations from these underlying trends and interpret spikes during election periods as responses to electoral incentives. We estimate:

$$\begin{aligned} y_{mt} = \delta_1 \text{Election Year}_t + \sum_{k=2}^{4} \delta_k \text{Quarter}_t^k \times \text{Election Year}_t \\ + \beta X_{mt} + \text{Year}_t + \text{Year}_t^2 + \gamma_{q(t)} + \gamma_m + \varepsilon_{mt} \end{aligned} \tag{2}$$

where y_{mt} is the outcome variable of interest, the tubal ligation rate in municipality m in month t (per 100,000 female residents of reproductive age). The coefficient δ_1 captures the deviation in the tubal ligation rate during the first quarter (Q1) of municipal election years relative to nonelection years, after controlling for municipality fixed effects (γ_m), calendar quarter fixed effects ($\gamma_{q(t)}$), time trends (Year_t and Year_t^2), and time-varying covariates X_{mt}. The interaction terms δ_2, δ_3 and δ_4 capture how much the effects in the next three quarters of municipal election years depart from this Q1 effect; the total effect in quarter $k \geq 2$ is $\delta_1 + \delta_k$. This specification allows us to assess whether the electoral cycle intensifies over the course of the election year, relative to Q1 (the omitted reference quarter). Of particular interest is δ_3, which corresponds to the campaign period (July–September) immediately preceding the October election. If electoral incentives influence the provision of sterilizations, we would expect δ_3 to be large and positive, indicating a sharp increase in tubal ligations during the campaign quarter beyond any effect already present at the start of the election year. We report the total campaign-quarter effect, $\delta_1 + \delta_3$, at the bottom of each relevant table.

It is worth noting that because municipal elections in Brazil occur simultaneously across

all municipalities, a standard difference-in-differences design is infeasible: there is no untreated control group in any given election year. Instead, our identification relies on comparing outcomes across quarters within election years to the same quarters in nonelection years, leveraging the predictable timing of elections within the electoral calendar. Quarter fixed effects absorb seasonal patterns common to all years, while year trends account for secular changes in procedure rates, so that the estimated δ_k coefficients capture deviations specific to election-year quarters.

As an alternative to parametric time trends, we also implement year-pair fixed effects, where a year pair is defined as a municipal election year and the year immediately preceding it. This approach directly compares outcomes in adjacent years within the same municipality, flexibly accounting for municipality-specific time trends without imposing a functional form. Relative to the quadratic trend specifications, year-pair fixed effects also address several additional concerns: they absorb local economic shocks or changes in healthcare capacity that evolve gradually across municipalities, and they mitigate concerns about using a fixed population denominator to construct procedure rates, since the population of women of reproductive age in a given municipality is unlikely to shift meaningfully over a one-year window.

When examining within-year variation, our primary specifications restrict analyses to municipal election years and the year immediately preceding them.[37] This restriction has two key benefits. First, it enables us to focus on the dynamics of female sterilizations around municipal elections by excluding years when national elections are held. Second, it omits the post-election year, during which new administrations take office and governance transitions may introduce changes unrelated to electoral incentives.[38] By narrowing our comparison to the year immediately preceding an election — when the same local politicians are in office and seasonality likely remains unchanged — we isolate a clean control to sharpen identification of electoral cycles in sterilization rates. We show the robustness of results to the inclusion of all years in Appendix Table A6.

Our within-year identification strategy relies on the assumption that, absent electoral incentives, tubal ligation rates in election years would follow the same seasonal pattern as in the pre-election years used as controls. This assumption is supported by Figure 2a and Figure 2b, which show that tubal ligation rates in election years and pre-election years track each other closely in the first two quarters, and then diverge in the third quarter just before the election. Nevertheless, if politicians begin reallocating resources or signaling their intent to provide sterilizations before the election year itself, the pre-election year may not be entirely free of electoral influence. To the extent that such anticipatory behavior occurs, our estimates represent a lower bound on the true magnitude of electoral effects.

4 Results

Given the intriguing patterns observed in Figure 1 — as well as qualitative evidence of politicians using tubal ligations for political purposes — this section employs the empirical strategies described above to investigate electoral cycles in female sterilization.

4.1 Electoral Cycles in Female Sterilization: Annual Variation

In Table 1, we present estimates from several specifications based on Equation 1. As discussed above, this analysis examines annual-level variation in tubal ligation rates to discern whether municipal election years have systematically more female sterilizations. Column 1, which only controls for quadratic time trends, reveals that municipal election years are associated with significantly higher rates of tubal ligations. Specifically, the results indicate that the sterilization rate increases by 0.155, versus the monthly average rate during nonelection years of 2.1 procedures per 100,000 women of reproductive age. This finding, which is significant at the 1% level, suggests that the tubal ligation rate increases by 7.3% during election years. Column 2 shows that this estimate is nearly identical when controlling for the log of municipal GDP per capita, suggesting that the observed increase is not simply driven by cyclical variation in local economic activity. Column 3 includes municipality fixed effects to account for unobserved, time-invariant differences across municipalities. The estimated effect

of municipal election years increases slightly to 0.171 and remains statistically significant, providing additional evidence that the pattern is not driven by systematic differences in sterilization rates across municipalities.

[Table 1 about here]

To directly compare the effects of municipal and national elections, Column 4 includes both indicators, revealing which type of election year is more strongly associated with changes in female sterilization rates. We find that the increase in tubal ligation rates during municipal election years is double that of the increase during national election years. This specification controls for municipality fixed effects, log GDP per capita, and quadratic time trends. More specifically, the effects during municipal and national election years are 0.201 and 0.097, respectively — corresponding to 8.8% and 4.3% increases compared to the average rate during nonelection years (2.27). This pattern suggests that local politics may play a particularly important role in electoral cycles in healthcare provision.

One might be concerned that the observed increase in tubal ligations during election years merely reflects a broader rise in healthcare provision, rather than a specific increase in female sterilizations. To address this concern, Column 5 also controls for the total inpatient procedure rate (AIH rate), capturing general trends in service delivery. Taking into account the possibility that overall healthcare provision may itself respond to political cycles, conditioning on the AIH rate allows us to assess whether tubal ligations behave differently from other inpatient procedures. The results show that the effect of municipal election years remains strong and statistically significant even after accounting for broader shifts in inpatient care.[39] In contrast, the estimated effect for national election years falls substantially and is no longer statistically significant. This finding suggests that the increase in the tubal ligation rate during national election years observed in Column 4 stems from broadly higher healthcare provision in those years, rather than targeted provision of the procedure.

These results are robust to a range of alternative specifications. Recall from Section 2 that our primary specifications focus on tubal ligations performed on women residing

in municipalities with a hospital and exclude surgeries during cesarean births. In Appendix Table A3, we relax these restrictions and analyze the full universe of over one million publicly provided sterilizations — that is, any procedures conducted in any municipalities. The estimated effects remain similar in magnitude and significance, indicating that results are not driven by sample selection. In addition, we replicate the analysis using the count of tubal ligations as the dependent variable, rather than rates per 100,000 women of reproductive age. As shown in Appendix Table A4, results are robust, suggesting that findings do not depend on how the outcome is scaled. Finally, to investigate whether findings are driven by the first half of the sample period, we partition our dataset into two time periods (1998–2008 and 2009–2019) and replicate the specifications in Table 1. As shown in Appendix Table A5, electoral cycles in female sterilization are statistically significant at the 1% level in both periods.[40] While the absolute increase during election years is comparable across both periods, the proportional effect is larger in 1998-2008, reflecting the lower baseline sterilization rate during this earlier period when publicly funded tubal ligations were still expanding following legalization.

4.2 Electoral Cycles in Female Sterilization: Within-Year Variation

Thus far, our year-level analysis reveals a systematic increase in tubal ligations performed during municipal election years. By leveraging the high granularity of data from DATASUS, we now conduct more precise analyses of when sterilizations increase. In particular, we examine whether the excess provision of tubal ligations is concentrated in months immediately preceding an election, when the electoral incentives to deliver services are strongest.

As discussed above, we estimate deviations from underlying seasonal and long-term trends following Equation 2. As a baseline against which to compare deviations during municipal election years, we employ data from the year preceding each municipal election year. These pre-election years serve as a credible control group because they are subject to the same seasonal patterns, institutional dynamics, and governance structures as municipal election years, but without immediate electoral incentives.[41] By focusing on departures from expected

trends, we can detect whether the public provision of female sterilization intensifies in the months before municipal elections, as would be expected if local politicians are strategically allocating healthcare resources to influence voters.

Figure 2a compares the evolution of tubal ligation rates in years with and without municipal elections. The red dashed line illustrates the average monthly tubal ligation rate in pre-election years (i.e., when electoral incentives are muted). This red dashed line serves as a baseline for comparison purposes, and shows seasonality in female sterilizations that is accounted for in regression analyses below. The blue solid line represents the average tubal ligation rate during municipal election years. For the first half of the year, both lines exhibit similar trajectories. However, beginning around July, the lines diverge: sterilization rates during election years rise, peaking in August and September — immediately before the election on the first Sunday in October — whereas rates in pre-election years remain relatively flat. This pattern suggests a sharp increase in the provision of tubal ligations aligned with the electoral calendar.

[Figure 2 about here]

We extend this graphical analysis in Figure 2b, which employs a regression framework to estimate differences in the public provision of tubal ligations.[42] As with Figure 2a, the results show that in the first half of election years, deviations from seasonal norms are generally insignificant. But in the third quarter, starting in July, we see a rise in the tubal ligation rate, which persists until the election. At the peak in August, we estimate that there are 0.840 additional procedures per 100,000 women of reproductive age, corresponding to a remarkable 42.0% increase over the expected level in August absent electoral effects.

Beyond these graphical analyses, we formally quantify the effects of electoral cycles in Table 2. In particular, we implement the specification in Equation 2 described in Section 3.2. Column 1 reports estimates from a baseline specification that includes municipality fixed effects, quarter fixed effects, and quadratic year trends. As explained above, the coefficient δ_1 captures the difference in the tubal ligation rate in Q1 of election years relative to nonelection

years, while δ_2, δ_3 and δ_4 capture how much the effects in the next three quarters of election years depart from this Q1 effect; the total effect in quarter $k \geq 2$ is $\delta_1 + \delta_k$. In the first half of the year, when electoral incentives are weaker, neither the Q1 effect (δ_1) nor the total Q2 effect ($\delta_1 + \delta_2$) is statistically distinguishable from zero. By contrast, in Q3 of municipal election years, the total effect ($\delta_1 + \delta_3$, reported at the bottom of Table 2) indicates that the tubal ligation rate rises by 0.610 procedures per 100,000 women of reproductive age, a substantial increase that is significant at the 1% level. This third-quarter effect corresponds to a 30.5% increase in female sterilizations, relative to nonelection years.

[Table 2 about here]

These findings are robust to a range of specifications. Column 2 controls for the log of GDP per capita to account for changes in local economic conditions; estimated coefficients are stable and remain significant. Column 3 shows that findings are similarly robust when also controlling for the total inpatient procedure rate (i.e., the AIH rate), which accounts for fluctuations in healthcare provision more broadly. To the extent that electoral cycles exist in other healthcare services (discussed below), this specification helps to isolate a differential effect on tubal ligations; the stability of estimates suggests that our findings do not merely reflect a general increase in hospital activity during election periods.

As a further robustness test, Columns 4 and 5 implement year-pair fixed effects as described in Section 3.2. This step addresses the possibility that quadratic year trends do not adequately capture underlying long-term trends in tubal ligations. A year pair refers to a municipal election year and the year immediately preceding it. As discussed above, by comparing adjacent years this approach not only more flexibly accounts for local time trends, but also mitigates potential concerns about municipality-specific shocks to healthcare capacity and about the use of a fixed population denominator. In the first two quarters of the year, we find higher female sterilization rates in municipal election years, which is consistent with the overall upward trend in procedure rates over time. Then, as with other specifications, we observe a statistically significant surge in female sterilizations in the third quarter of

election years.[43] As an additional robustness test, Appendix Table A8 shows that findings in Table 2 remain significant when replicating the analysis using the count of tubal ligations as the dependent variable, rather than rates per 100,000 women of reproductive age.

While not our primary focus, it is worth noting that in Columns 1-2 as well as in Figure 2b, we observe a decrease in the tubal ligation rate in the final quarter of municipal election years. This pattern is consistent with a drop in electoral incentives following the October election, as the political returns to service provision are lower after votes are cast. The decline in procedures during this period also reflects a broader contraction in healthcare delivery that typically follows municipal elections in Brazil. As shown in Column 3, the effect in the fourth quarter becomes small and statistically insignificant once we control for the total inpatient procedure rate, suggesting that the decline in tubal ligations is not disproportionately larger than the general post-election decrease in service provision.[44]

Stepping back, quantitative analyses suggest that female sterilizations surge during election campaigns. Employing annual variation, we observe that female sterilizations are substantially greater during municipal election years. More rigorous analyses using within-year variation suggest that this spike in tubal ligations is concentrated in the three months just before municipal elections are held in Brazil.

4.3 Heterogeneity of Electoral Cycles

Within Brazil, in what contexts do we observe greater electoral cycles in female sterilization? One motivation for examining heterogeneity is that the broader literature on political budget cycles suggests politicians may be especially likely to increase their pre-election expenditures under some conditions — a phenomenon termed "conditional budget cycles."[45] For example, Shi and Svensson (2006) show that these cycles are greater in developing countries, and develop a model to explain why: they argue that cycles are less pronounced when voters are more informed, and more pronounced when politicians can more easily obtain rents from office. Likewise, studies on clientelism also point to heterogeneity: for instance, the *quid pro quo* exchange of benefits for votes is most prevalent in developing countries, and

within countries, politicians disproportionately distribute clientelist handouts to poor citizens (Hicken, 2011; Stokes et al., 2013).[46] While such literature motivates our examination of heterogeneity, we emphasize that the findings below do not definitively point to underlying mechanisms; for example, patterns of heterogeneity may instead reflect greater baseline unmet demand, weaker administrative oversight, weaker fiscal institutions, and/or greater discretion in scheduling.

To assess heterogeneity, we partition the dataset and conduct regressions analogous to those above for each subgroup.[47] Thus, Table 3 employs within-year variation and we again focus on the third quarter of municipal election years — corresponding to the campaign period, when electoral incentives are strongest. All specifications include municipality and quarter fixed effects, as well as quadratic year trends.

[Table 3 about here]

We first investigate regional heterogeneity. While clientelism and other forms of non-programmatic politics are observed across Brazil, a wide literature has emphasized their prevalence in the Northeast region of the country.[48] Poverty is especially concentrated in the Northeast: the region accounts for 27% of Brazil's population, but 44% of Brazil's poor and 55% of its extreme poor (IBGE, 2023). To explore whether electoral cycles are amplified in Northeast Brazil, Columns 1 and 2 replicate analyses above while partitioning the dataset into tubal ligations performed within versus outside the Northeast region. Results suggest that the surge in female sterilization during elections is indeed especially pronounced in the Northeast region, where the tubal ligation rate increases by 69.3% in the three months prior to a municipal election.[49] This increase is over four times the magnitude of the increase in other regions, which experience a 15.1% increase in procedures during the campaign period.[50] Both effects are significant at the 1% level. Whereas the Northeast region accounts for 22.9% of publicly provided female sterilizations during the study period, the region accounts for 66.8% of the total increase observed across Brazil during municipal elections.

Given that studies on both political budget cycles and clientelism suggest that these phenomena are especially prevalent in poorer contexts, we next investigate whether electoral cycles in tubal ligations are amplified in poorer municipalities. Columns 3-5 replicate analyses above while partitioning the dataset into terciles of municipal GDP per capita.[51] As expected, effects are strongest in municipalities in the bottom income tercile: the monthly tubal ligation rate spikes by 57.4% in the third quarter of municipal election years (significant at the 1% level).[52] The middle income tercile of municipalities exhibits a smaller increase of 24.2% in the third quarter of municipal election years, while the richest tercile exhibits a 17.8% increase (both significant at the 1% level).[53] This gradient suggests that electoral cycles in female sterilization procedures are amplified in poorer areas.

Beyond municipal characteristics, our dataset also provides limited information about patients, which sheds light on another key aspect of potential heterogeneity. We examine whether electoral cycles are especially pronounced for sterilizations of nonwhite women, who face significant health disparities in Brazil.[54] Columns 6-7 examine this heterogeneity by partitioning the data into tubal ligations performed on white vs nonwhite women.[55] The specifications show that in the third quarter of municipal election years, the tubal ligation rate for nonwhite women rises by 0.414 procedures per 100,000 nonwhite women of reproductive age (significant at the 1% level). By contrast, the tubal ligation rate for white women rises by a statistically insignificant 0.160 procedures per 100,000 white women of reproductive age. Because the baseline sterilization rate for nonwhite women is substantially greater than for white women, sterilizations increase by a similar percentage across race. Nevertheless, electoral cycles in female sterilization are concentrated among nonwhite Brazilians, who face substantial health disparities: a back-of-the-envelope calculation suggests that over 80% of the excess sterilizations during election campaigns are performed on nonwhite women.[56] These racial disparities persist even when restricting the analysis to the Northeast region (see Columns 8-9).

Given temporal trends observed in Figure 1, one might be concerned that our findings

only hold during the earlier part of our sample. While all analyses above undertake steps to mitigate such concerns (e.g., including quadratic time trends or year-pair fixed effects), we examine heterogeneity to counter the possibility that our findings only hold in the first half of our sample period. As shown in Appendix Table A13, electoral cycles in female sterilization are statistically significant in both 1998–2008 and 2009–2019.[57]

In sum, these heterogeneity analyses suggest that electoral cycles in female sterilizations are especially amplified in poor municipalities and in the Northeast region. Moreover, electoral cycles in female sterilization are concentrated among nonwhite Brazilians, who face substantial health disparities. Electoral cycles in female sterilization are significantly observed in both the earlier and later years of our sample period.

4.4 Electoral Cycles in Other Hospitalizations

Do these electoral cycles in female sterilizations merely reflect broader increases in public healthcare provision during campaigns? Reassuringly, analyses above already show that findings are robust when controlling for any electoral cycles in total hospitalizations. But the question remains whether electoral cycles in tubal ligations are unique. For example, given that the literature on political budget cycles suggests that politicians often increase their provision of targetable benefits that are highly valued by voters (Brender & Drazen, 2005; Drazen & Eslava, 2010), we examine below whether similar patterns are observed for other elective surgeries that have substantial waiting lists. More specifically, we focus on the three types of elective surgeries with the largest number of patients awaiting treatment in Brazil's public healthcare system (hernia repairs, gallbladder removals, and cataract surgeries),[58] as well as vasectomies, the male analog of tubal ligations. We also examine emergency procedures (appendectomies and heart attack treatments) as falsification tests, since their timing is dictated by medical urgency rather than administrative discretion.

Before examining other surgeries, we document the extent of electoral cycles in hospitalizations overall. More specifically, Column 1 of Table 4 now employs the AIH rate as a dependent variable instead of as a control variable; it reflects how many inpatient procedures

were provided by the public healthcare system per 100,000 residents. As shown, we do find evidence of electoral cycles in the broader provision of healthcare, but the magnitude is quite small — in the third quarter, when electoral incentives are strongest, there is only a 0.8% increase in the overall hospitalization rate. This finding stands in sharp contrast to the 30.5% increase in tubal ligations during the same period, highlighting that while political cycles may influence general healthcare activity, female sterilizations exhibit a disproportionately large response.

[Table 4 about here]

Second, we examine whether there are electoral cycles in vasectomies. Male sterilization through vasectomies, though less invasive than female sterilization and not requiring general anesthesia, continues to be considerably less prevalent in Brazil.[59] Column 2 examines the vasectomy rate and finds substantial evidence of electoral cycles: in the third quarter of municipal election years, the rate increases by 11.7%. This effect is statistically significant at the 5% level. Nevertheless, it deserves emphasis that the effect for female sterilizations is nearly triple in proportional terms (30.5% versus 11.7%), and even more so in absolute terms given their substantially larger baseline rate.

Third, we examine electoral cycles in the provision of other surgeries. Columns 3-5 focus on the three elective procedures with the largest queues in Brazil's public healthcare system: hernia repairs, gallbladder removals, and cataract surgeries. Although these procedures may be medically necessary, they are generally non-urgent and therefore may be especially subject to scheduling delays and rationing. We find evidence of increased provision for two of these three elective surgeries during municipal election campaigns, with statistically significant increases of 10.4% and 5.8% for hernia repairs and gallbladder removals (Columns 3 and 4, respectively). By contrast, no significant increase is observed for cataract surgeries (Column 5). As a falsification test, we examine electoral cycles in two key nonelective surgeries (appendectomies and heart attack treatments). As shown in Columns 6-7, we do not detect any increased provision of these nonelective surgeries during campaigns. These procedures

require timely and often emergency care, perhaps leaving little scope for strategic influence by political actors.

Taken together, these results suggest that while electoral cycles in female sterilizations are especially large, similar patterns are observed for some other healthcare procedures. In proportional terms, the campaign-period increase for tubal ligations (30.5%) is the largest among all procedures examined, far more than that of vasectomies (11.7%), hernia repairs (10.4%), and gallbladder removals (5.8%).[60] Elective procedures with long queues offer local officials an opportunity to deliver targetable and highly valued benefits in the months leading up to an election. By contrast, the provision of healthcare overall exhibits minimal cyclicality.

5 Conclusion

The present study explores electoral cycles in female sterilization, the most prevalent contraceptive method in the world. We focus on the case of Brazil, a country where many women have substantial unmet demands for sterilization. As our findings suggest, in this context where many eligible women wait in lengthy queues for tubal ligations, politicians distort the timing of these surgeries for electoral purposes.

Given substantial qualitative evidence of this phenomenon, we analyze a panel dataset of the universe of publicly funded tubal ligation surgeries across Brazilian municipalities in 1998-2019. Regressions demonstrate that the incidence of female sterilization increased 8.8% during municipal election years. The increase in tubal ligation surgeries is especially large during electoral campaigns: female sterilizations surged by 30.5% in the three months immediately preceding municipal elections. Findings are similar when adjusting for the overall provision of inpatient procedures, which increase less than 1% during municipal elections. While our study focuses on female sterilization, we also demonstrate significant but considerably smaller electoral cycles in vasectomies. Female sterilizations also have proportionally larger electoral cycles than do other elective surgeries with long queues, and no cycles are detected when investigating emergency surgeries. Electoral cycles in female sterilization are

concentrated among nonwhite Brazilians and are amplified in poor municipalities and in the Northeast region.

An important avenue for future research involves the mechanisms underpinning these electoral cycles. Our findings are consistent with the broad literature on political budget cycles, which suggests various reasons why politicians across the world often increase fiscal expenditures before elections, such as signaling competence or demonstrating policy preferences (e.g., Nordhaus, 1975; Rogoff & Sibert, 1988). Yet they are also consistent with Hanusch & Keefer's (2014) argument that electoral cycles may stem from clientelism; i.e., in line with some qualitative evidence discussed above, observed patterns may involve the *quid pro quo* exchange of sterilizations for political support. A key next step is to unpack how and why Brazilian politicians increase female sterilizations during campaigns. While our use of SUS data is advantageous in that it provides the universe of all publicly provided tubal ligations in Brazil, our dataset does not facilitate testing for clientelism (e.g., examining if a mayor's partisans are disproportionately sterilized) because public health authorities do not report political information about patients. If the Brazilian government authorizes linking records, additional data could shed light on whether women's partisanship or voting behavior render them more likely to receive sterilizations during election campaigns (*ceteris paribus*).[61] Distinguishing between potential mechanisms could have important policy implications, as the appropriate institutional response may differ depending on how and why the provision of sterilizations increases during electoral periods.

Such research on underlying mechanisms could also shed light on an important open question: why are these specific procedures — tubal ligations — chosen at such high rates for electoral purposes? Among various possibilities, future empirical and theoretical research should explore whether politicians who increase the provision of healthcare during elections prioritize highly demanded services that are relatively less costly and/or disproportionately serve younger voters with longer electoral horizons. In Brazil, many elective surgeries are in high demand: as noted above, 68% of respondents in a representative Federal Council of

Medicine survey (CFM, 2018) indicated that surgeries were "difficult" or "very difficult" to obtain; this accessibility was the second worst of 14 categories of SUS services (after specialty consults). Among the elective surgeries with the largest queues, politicians may prioritize relatively cheaper (or less complex) operations for electoral purposes, which would allow them to distribute a greater number of procedures within a given budget. Lower-cost surgeries may also attract less oversight, reducing the risk of legal consequences. In Brazil, the cost of tubal ligations performed by SUS was 31% to 54% cheaper than the procedures with the top three queue sizes evaluated above (i.e., hernia, gallbladder and cataract surgeries).[62] Furthermore, politicians may prioritize surgeries performed on younger voters, who are expected to vote in a greater number of future elections and thus have the potential to yield more long-term votes for their party. In Brazil, the average age of tubal ligation patients (32 years) was considerably lower than that of patients receiving cataract surgeries (68 years), gallbladder removals (47 years), and hernia operations (46 years).[63] These possibilities remain untested, but suggest that this line of exploration warrants greater attention.

Given that our findings demonstrate pronounced electoral cycles in female sterilization, welfare effects also deserve further examination. Identifying these effects may be complicated by the fact that electoral cycles may involve some combination of net expansion (i.e., performing sterilizations that otherwise would not have been performed) and intertemporal reallocation (i.e., shifting the timing of existing sterilizations). Expanding access to reproductive health may improve welfare, depending on how that expansion affects other service provision. For example, it is important to consider whether politicians shift resources from other services (e.g., preventative care) if sterilizations are considered more electorally valuable.[64] The welfare effects of intertemporal reallocation also require careful examination. On the one hand, elections may accelerate access to sterilizations for women who have already endured long queues for those procedures. But if politicians fund increased sterilizations during campaigns by rationing their provision outside of election periods, both reproductive health and downstream health outcomes could suffer. For instance, if accelerating steril-

izations comes at the cost of reduced provision directly after elections (also consistent with our findings), women seeking sterilizations in the post-electoral period may face even longer queues. Furthermore, an important consideration is how electoral cycles might affect *which* women receive sterilizations. If politicians shift the profile of sterilization recipients during elections, reproductive health effects might disproportionately affect some subpopulations. Overall, assessing the welfare implications of our findings is an important direction for future research.

Investigating such questions would shed light on the broader consequences of electoral cycles in female sterilization. Since millions of women are sterilized each year across the globe, further work should also explore whether similar electoral cycles are observed beyond Brazil. While a rigorous evaluation of external validity would require testing in other countries, numerous conditions may render such patterns more likely. For example, the literature on political budget cycles suggests they may be more likely in less developed countries with greater rent-seeking activities and less-informed voters (Shi & Svensson, 2006), where fiscal transparency is low (Alt & Lassen, 2006), and where elections are competitive (Tufte, 1978). More specifically with regard to tubal ligations, electoral cycles may potentially be more likely in contexts where: the public sector provides most sterilizations; there is unmet demand and lengthy queues for public sterilizations; many women struggle to obtain sterilizations privately (e.g., due to poverty or rurality); laws provide politicians substantial discretion in public health expenditures; monitoring and enforcement of such laws are weak; or an underdeveloped judicial system fosters impunity among politicians. More broadly, assessing whether politicians distort the public provision of contraceptive methods is a critical task, not least because family planning is widely understood to be a basic human right.

Notes

[1]For example, see the 1968 International Conference on Human Rights, the UNFPA's "Strategy for Family Planning, 2022-2030," and the WHO website (https://www.who.int/news-room/fact-sheets/detail/family-planning-contraception).

[2]According to the United Nations (2022), India accounts for much of the prevalence of female sterilization; the procedure's share among married contraceptive users is 59% in India, 53% in El Salvador, 48% in Mexico, 45% in the Dominican Republic, and 40% in Nicaragua. Male sterilization's share is 2% worldwide.

[3]For example, see Alonso (2020) and Patel (2017), as well as the interagency statement by the United Nations, WHO and other organizations (OCHR et al., 2014).

[4]Researchers have found strong evidence of political budget cycles in various contexts including China (Guo, 2009), Egypt (Blaydes, 2010), India (Khemani, 2004), Russia (Akhmedov & Zhuravskaya, 2004), and across Latin America (Ames, 1987). Although some studies on developed countries also identify political budget cycles (e.g., Reid, 1998; Galli & Rossi, 2002), Shi & Svensson (2006) argue that the amplitude of cycles is greater in poor countries. For excellent reviews on this topic, see Franzese (2002) or Aaskoven & Lassen (2017).

[5]Among Brazilians hospitalized in the past year, the vast majority of poorer Brazilians were hospitalized by SUS in their last visit: 95% for those earning under 1/4 the minimum salary, 90% for those earning 1/4 to 1/2 times the minimum salary, and 77% for those earning 1/2 to 1 times the minimum salary (IBGE, 2020). By contrast, the corresponding percentage was just 6.8% for Brazilians earning over 5 times the minimum salary, as wealthier citizens often employ private health plans to complement or circumvent SUS.

[6]Moreover, 84% of Brazilians who did not complete elementary school rely exclusively on SUS, versus 32% of those with university degrees.

[7]Municipal elections occur concurrently across all Brazilian municipalities; state and national elections follow two years later. Mayors can hold office for two consecutive terms (later reelection is permissible). Councilors serve as legislators at the municipal level, are elected by open-list proportional representation, and do not face term limits.

[8]Instituto de Pesquisa Econômica Aplicada (IPEA), "Gasto Público em Saúde," accessed February 27, 2026, https://www.ipea.gov.br/portal/beneficiometro/beneficiometro-artigos/saude/gasto-publico-em-saude

[9]By focusing on female sterilizations through December 2019, we circumvent their sharp decrease due to the COVID-19 pandemic, as well as their subsequent increase due to a 2022 legal reform. Among other changes, Law 14,443 in 2022 lowered the age requirement from 25 to 21 years, thereby substantially increasing eligibility for tubal ligations.

[10]The CFM conducted this representative survey of 2,087 Brazilians with Datafolha.

[11]Based on these data, the CFM filed a complaint with the Chamber of Deputies about long waiting lists for elective surgeries.

[12]Data on the waiting list (in June 2017) and surgeries performed (in all of 2017) in Minas Gerais are from Federal Council of Medicine (2017) and DATASUS (2025), respectively. In the city of Rio de Janeiro in 2017, 3,184 women awaited sterilizations (see "A fila do desamparo na rede municipal," *O Globo*, 1/8/2017). More systematic data on waiting lists for female sterilization are unavailable during the study period.

[13]For example, academic studies include Berquó & Cavenaghi (2003), Osis et al. (2009), Caetano (2014) and Barbosa (2018). News articles include: "Fila da laqueadura dura até 2,5 anos," *Diário do Grande ABC*, 12/12/2005; "Moradora da Capital fica dois anos à espera de laqueadura," *Zero Hora*, 5/4/2015; and "A fila do desamparo na rede municipal," *O Globo*, 1/8/2017.

[14]The Ministry of Health's program description is available here: https://www.gov.br/saude/pt-br/composicao/saes/drac/pnrf.

[15]The Conclusion discusses how enhanced data availability could facilitate testing for clientelism. More broadly, clientelism — the contingent (*quid pro quo*) exchange of benefits for political support — has long precedent and continues in Brazil (e.g., Bobonis et al., 2022; Diniz, 1982; Hagopian, 1996; Nunes Leal, 1949; Weyland, 1996). The phenomenon is particularly rife during municipal elections, and city councilors often serve as clientelist brokers on behalf of allied mayors (Nichter, 2018).

[16]"Reportagem Complica a Candidatura de um Médico em Fortaleza," *Diário do Pará*, 8/16/1990.

[17]"Prática de compra de votos é investigada em mais de mil municípios brasileiros," *Fantástico*, 8/6/2012 at https://globoplay.globo.com/v/3614370/.

[18]This message was advertised to voters in 2010 by *Movimento de Combate à Corrupção Eleitoral*, a leading umbrella network of civil society organizations against clientelism.

[19]The following news article quoted ex-federal deputy Raul Jungmann (from Pernambuco in Northeast Brazil): "Transparência é mau negócio para deputados governistas," *O Estado de São Paulo*, 1/30/2011.

[20]Information in this paragraph is from the following publication: "Condenado deputado federal paraense por crime de esterilização irregular," Notícias de Supremo Tribunal Federal, 9/8/2011. This politician was convicted seven years after buying votes as a mayoral candidate, by which time he had been elected federal deputy.

[21]Information in this paragraph is from the following court decision: Tribunal Regional Eleitoral de Rondônia, Acordão No. 97 de 17/4/2008, Processo No. 88 - Classe 7.

[22]Information in this paragraph is from the following court decision: Tribunal Regional Eleitoral de Goiás, Recurso No. 5718, Protocolo 3392052008, 5/18/2009.

[23]Information in this paragraph is from the following sources: Tribunal Regional Eleitoral do Rio de Janeiro, 8/10/2012, Document No. 64-53.2010.6.19.015; "TRE-RJ Mantém Candidatura de Ex-Prefeito Cassado em 2005," *O Globo*, 9/11/2012; and "Hospital e Investigado por Trocar Cirurgias por Votos," *Correio do Brasil*, Ano XI, Número 4300, 9/9/2010.

[24]"Hospital e Investigado por Trocar Cirurgias por Votos," *Correio do Brasil*, Ano XI, Número 4300, 9/9/2010.

[25]More specifically, his 2010 election for substitute state deputy was vacated. The candidate's brother received a monetary fine for his involvement.

[26]Information in this paragraph is from the following court decision: Tribunal Superior Eleitoral, Decisão Monocrática de 31/10/2023, Document No. 060046111.

[27]Fieldwork was conducted by the author in Bahia and Pernambuco, including 55 elite interviews in 2008-2009 and 16 elite interviews in 2012. These interviews included mayors, former mayors, city councilors, vice-mayors, party heads, and heads of social services. During the same fieldwork, similar views about a broad array of services were also frequently conveyed in interviews with 61 citizens. For example, one citizen explained: "They save part of the funds and they do it in the last year, so, one year before the election, they do everything ... to show 'I did it.'" (Author's interview in a Bahian municipality with 10,000 citizens, October 16, 2008.)

[28]Author's interview in a municipality with 10,000 citizens on November 26, 2008.

[29]Author's interview in a municipality with 30,000 citizens on December 4, 2008.

[30]"Sesau envolvida na eleição" and "Enfermeira diz que prática é antiga," *Jornal do Comercio*, 11/9/1986.

[31]Analyses below employ data on the race of patients, which is only available from 2008 onward.

[32]To ensure a balanced panel, our sample excludes the 1% of Brazilian municipalities created after 1998.

[33]See the following Ministry of Health website: https://www.gov.br/saude/pt-br/sus/responsabilidades-dos-entes-que-compoem-o-sus.

[34]To identify municipalities with hospitals, we rely on data from the National Registry of Healthcare Facilities (*Cadastro Nacional de Estabelecimentos de Saúde* or CNES). We include municipalities that have at least one "General Hospital" or "Specialized Hospital," as these have the necessary surgical infrastructure to perform female sterilizations. In our data, municipalities with hospitals account for 98.9% of tubal ligations performed locally. To ensure a balanced panel, we use facility counts from 2005, the earliest year that the CNES registry is available.

[35]Another advantage of focusing on sterilizations of women who reside within the municipality involves measurement. For the denominator of our key dependent variable — the tubal ligation rate — we can employ census bureau data about the number of women of reproductive age who reside in each municipality.

[36]In municipalities with over 200,000 voters — a criterion met by less than 2% of Brazilian municipalities — a runoff election is also held several weeks later (on the last Sunday of October) if no mayoral candidate wins an absolute majority in the first round.

[37]By contrast, our first empirical strategy above includes all years to document that electoral cycles in tubal ligation rates are stronger in municipal election years than in national election years. As an additional robustness check, Appendix Table A2 shows that results are similar when employing year-level variation and restricting to municipal election years and the year immediately preceding them.

[38]This restriction also mitigates concerns about post-election changes in service delivery driven by new leadership. For instance, bureaucratic turnover following municipal elections has been shown to reduce healthcare provision (Toral, 2024). Including the year after elections could bias our estimates by contaminating the control group with governance-driven shifts in healthcare provision, thereby conflating post-electoral adjustments with the absence of pre-electoral strategies.

[39]As discussed in Section 4.4 below, electoral cycles in AIHs are significant but far smaller than those observed for sterilizations.

[40]A formal test of equality, based on a fully interacted model that interacts the election-year indicator with a post-2009 dummy, fails to reject the null hypothesis that the election-year effect is equal across the two periods ($F(1, 2997) = 0.53$, $p = 0.47$).

[41]To the extent that service provision in the pre-election year is also influenced by electoral considerations, the estimates reported here may represent a lower bound on the true effect.

[42]More specifically, we present estimates from a regression framework identical to Equation 2, except that we estimate monthly rather than quarterly deviations in the tubal ligation rate. The specification includes municipality and calendar month fixed effects. We plot the total effect for each given month.

[43]This specification estimates the difference in the sterilization rate for a given quarter in a municipal election year relative to the same quarter in the preceding year. Unlike all other specifications in Tables 1 and 2 that control for quadratic time trends, this approach does not adjust for projected year-on-year growth. Given the upward trajectory in tubal ligations over time (see Figure 1), for Columns 4 and 5, we expect higher rates in municipal election years than in preceding years, even without electoral incentives. Observe that when comparing the Q3 effect to the Q1 and Q2 effects, the relative difference is large and comparable in magnitude to Columns 1-3.

[44]In Table A14 of the Online Appendix, we show that this post-election drop in tubal ligations persists for approximately nine months. Figure A1 illustrates that this drop is driven by municipalities where incumbents are not reelected, consistent with evidence from Toral (2024), who documents that turnover disrupts healthcare delivery.

[45]For a literature review, see Aaskoven & Lassen (2017).

[46]Scholars have posited numerous reasons why clientelism targets the poor. Most prominently, the diminishing marginal utility of income suggests the poor place greater value on material benefits than on political preferences (Dixit & Londregan, 1996; Stokes, 2005). Researchers also point to the role of risk aversion (Desposato, 2007; Stokes et al., 2013) and time preferences (Kitschelt & Wilkinson, 2007; Scott, 1969).

[47]More specifically, we replicate Column 2 in Table 2, partitioning the dataset by the indicated subgroup.

[48]For example, see Nunes Leal (1949); Hoefle (1985); Roniger (1987); Bernal (2001); Bursztyn & Chacon (2013) and Moraes (2015).

[49]The tubal ligation rate increases by 1.382 monthly procedures per 100,000 females of reproductive age, compared to a baseline rate of 1.992 during the same quarter in nonelection years.

[50]The tubal ligation rate elsewhere increases by 0.303, compared to a baseline rate of 2.001.

[51]Given that most municipalities in the Northeast region are relatively poor, this finding overlaps with the regional results reported above.

[52]In this quarter, there is a 0.944 increase in monthly tubal ligations per 100,000 women in the poorest third of municipalities — relative to the 1.643 baseline rate in the same quarter of nonelection years.

[53]As shown, there is a 0.446 (0.446) increase in monthly tubal ligations per 100,000 women in the middle (top) tercile of municipalities — relative to the 1.837 (2.508) baseline rate in nonelection years.

[54]Among numerous reasons for examining such heterogeneity, recent quantitative research suggests that nonwhite citizens in Latin America may be disproportionately targeted with clientelist benefits during campaigns, even when controlling for their level of income (Johnson, 2020). It should be noted that DATASUS only provides information about race starting in 2008, and racial data are missing in subsequent years for approximately 40% of patients.

[55]Following the literature in Brazil, these two racial groups are based on the five categories provided by DATASUS. We use the "white" (*branca*) category in DATASUS, and aggregate the other four categories as "nonwhite": "black" (*preta*), "brown" (*parda*), "Asian" (*amarela*), and "indigenous" (*indígena*).

[56]As shown in Table A10 of the Online Appendix, in the average municipality there are an additional 0.0668 tubal ligations per month performed on nonwhite women in the third quarter of municipal election years. For white women, this figure is 0.0166 procedures per month.

[57]Furthermore, a formal test of equality, based on a fully interacted model that interacts all election-year quarter indicators with a post-2009 dummy, fails to reject the null hypothesis that the campaign-quarter effect is equal across the two periods ($F(1, 2999) = 1.09, p = 0.29$). Appendix Table A5 shows that year-level findings also hold during both periods.

[58]In the CFM study discussed above, these top three elective procedures collectively accounted for a third of the 904,000 citizens identified as in the queue for SUS surgeries (CFM, 2017a, 2017b). According to a presentation by the Ministry of Health's "National Queue Reduction Program," these three surgeries continued to have the largest waiting lists in early 2024, followed by tubal ligations (see here: https://www.gov.br/saude/pt-br/centrais-de-conteudo/apresentacoes/2024/apresentacao-pnrf-e-pmae).

[59]"Número de laqueaduras feitas pelo SUS mostra que responsabilidade do planejamento familiar recai mais sobre a mulher," *Jornal Nacional*, 4/10/2025. As noted in Section 2, our study examines Hospital Admission Authorizations (AIHs); as such, we analyze vasectomies performed as inpatient procedures (i.e., two-thirds of all SUS vasectomies in 2000-2019.

[60]We consider proportional comparisons to be most informative, given the wide variation in baseline rates across procedure types.

[61]For example, one approach would construct an individual-month panel dataset on sterilizations covering the universe of all eligible women that includes a partisanship variable (e.g., by linking the Brazilian government's census, health and party membership data); estimating the interaction of a partisan dummy and election quarter would test the hypothesis that partisans are disproportionately likely to be sterilized during electoral periods. An alternative approach would implement the same test but at the municipality (not individual) level, by linking our municipality-month panel to party membership data to construct counts of partisans and nonpartisans sterilized per municipality-month; to test the same hypothesis, one could employ within-year specifications as above to estimate whether there is a differential increase in partisan sterilizations in election periods relative to nonpartisan sterilizations. Similar analyses using electoral data on past turnout could test for turnout buying (Nichter, 2008), which would provide sterilizations to boost electoral participation. These approaches are by no means exhaustive but illustrate how further analyses could explore the mechanisms underlying observed patterns.

[62]These differentials reflect total costs in 2019; percentage differentials in 2010 are similar. More specifically, in 2019 (2010), the total SUS cost in AIHs of each tubal ligation was R$403 (R$346), compared to R$581 (R$485) for hernia repairs, R$784 (R$559) for cataract surgeries, and R$875 (R$774) for gallbladder removals. For vasectomies, the cost was R$378 (R$319).

[63]These ages reflect 2019 operations; the average ages in 2010 were 31, 67, 46 and 40. Vasectomy patients had an average age of 36 years in 2010 and 38 years in 2019.

[64]Rogoff (1990) develops a model that predicts politicians shift expenditures to goods visible to voters before elections. And with regards to clientelism, studies suggest that clientelist politicians underprovide non-targeted benefits and over-provide targeted benefits (Keefer, 2007; Keefer & Vlaicu, 2008).

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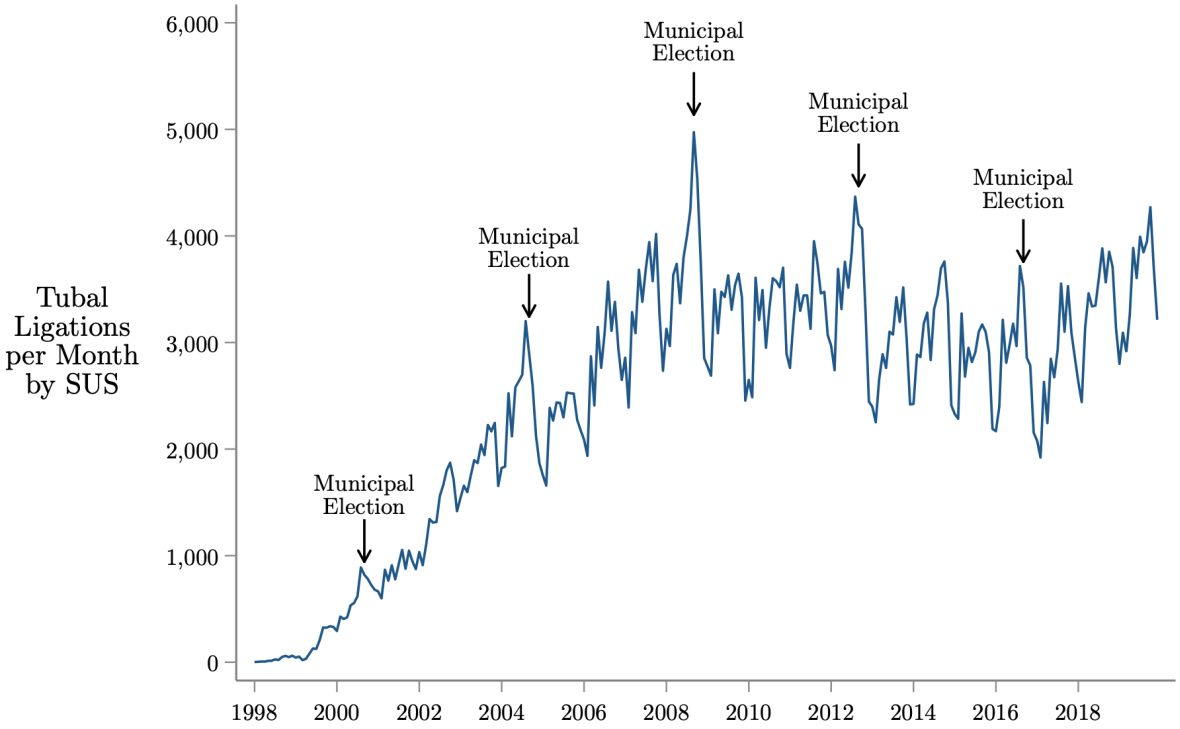
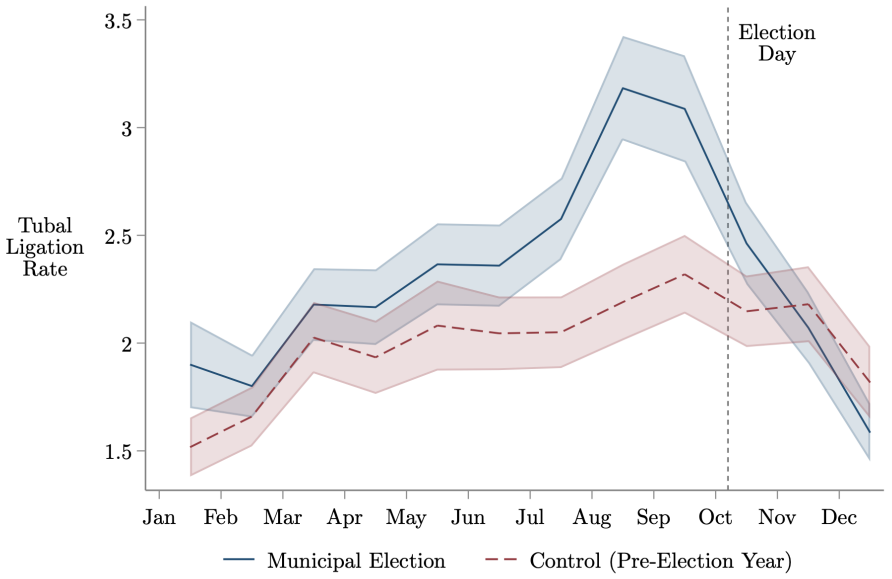
Figure 1: Tubal Ligation Procedures and the Timing of Municipal Elections



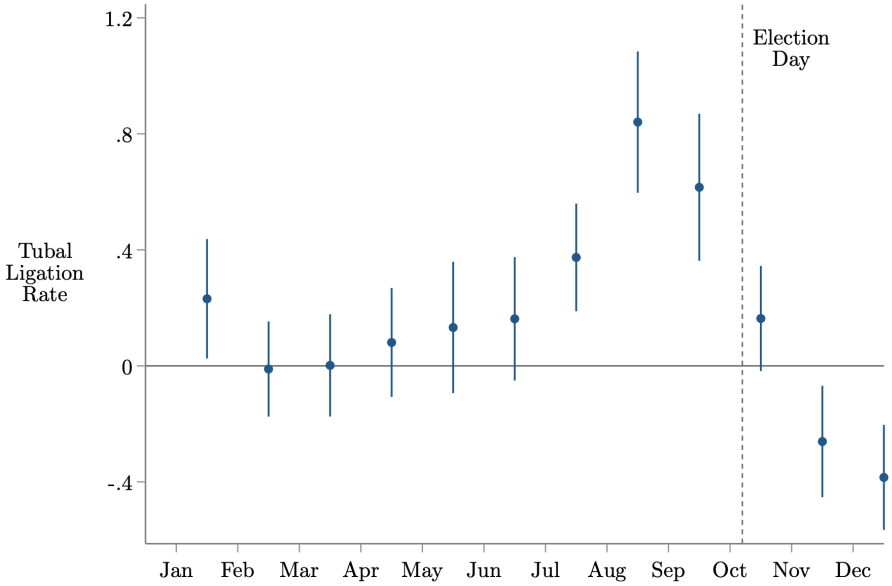
Notes: This figure shows the total number of tubal ligations performed each month through Brazil's public health system (SUS) between 1998 and 2019. Figure includes all tubal ligations, except surgeries performed during cesarean births. Municipal elections, which occur in October every four years, are shown.

Figure 2: Electoral Cycles in the Provision of Tubal Ligations

(a) Tubal Ligation Rates, Municipal Elections vs. Prior Year



(b) Estimated Differences in Tubal Ligation Rate



Notes: Panel A plots the mean monthly tubal ligation rate during municipal election years (2000, 2004, 2008, 2012, 2016; solid blue line) and the corresponding pre-election years (1999, 2003, 2007, 2011, 2015; dashed red line). Shaded regions represent 95% confidence intervals. The tubal ligation rate is defined as the number of procedures performed per 100,000 women of reproductive age (15–49) residing in the municipality. The sample is restricted to procedures performed in a patient's municipality of residence and to municipalities with at least one hospital. Panel B plots deviations from underlying trends in the tubal ligation rate, estimated using a regression framework analogous to Equation 2, except with monthly rather than quarterly deviations. The specification includes municipality fixed effects and quadratic time trends. 95% confidence intervals are shown, constructed using standard errors clustered at the municipality level.

Table 1: Electoral Cycles in Tubal Ligation Rates (Year-Level Variation)

	(1) Tubal Ligation Rate	(2) Tubal Ligation Rate	(3) Tubal Ligation Rate	(4) Tubal Ligation Rate	(5) Tubal Ligation Rate
Municipal Election Year	0.155***	0.155***	0.171***	0.201***	0.228***
	(0.0461)	(0.0462)	(0.0464)	(0.0448)	(0.0451)
Log GDP per Capita		0.729***	-0.268	-0.266	-0.192
		(0.121)	(0.269)	(0.269)	(0.263)
National Election Year				0.0966**	0.0591
				(0.0389)	(0.0387)
Total Procedure (AIH) Rate					0.00345***
					(0.000313)
Municipality FE	No	No	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Year Squared	Yes	Yes	Yes	Yes	Yes
Control Mean	2.137	2.270	2.270	2.270	2.270
Observations	792,000	755,484	755,484	755,484	755,484

Notes: This table reports estimates of the relationship between municipal election years and tubal ligation rates, as specified in Equation 1. Observations are at the municipality-month level. The outcome is the number of tubal ligations per 100,000 women of reproductive age (15–49) residing in the municipality. The sample is restricted to procedures performed in a patient's municipality of residence and to municipalities with at least one hospital. Standard errors clustered at the municipality level are shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Electoral Cycles in Tubal Ligation Rates (Within-Year Variation)

	(1) Tubal Ligation Rate	(2) Tubal Ligation Rate	(3) Tubal Ligation Rate	(4) Tubal Ligation Rate	(5) Tubal Ligation Rate
δ_1: Municipal Election Year	0.0741	0.0723	0.0662	0.226***	0.238***
	(0.0665)	(0.0667)	(0.0666)	(0.0659)	(0.0699)
δ_2: Municipal Election Year × Q2	0.0512	0.0502	0.0555	0.0512	0.0556
	(0.0760)	(0.0761)	(0.0760)	(0.0760)	(0.0760)
δ_3: Municipal Election Year × Q3	0.536***	0.536***	0.526***	0.536***	0.526***
	(0.0999)	(0.0999)	(0.0998)	(0.0999)	(0.0998)
δ_4: Municipal Election Year × Q4	-0.235**	-0.235**	-0.146	-0.235**	-0.145
	(0.0961)	(0.0962)	(0.0955)	(0.0961)	(0.0955)
Log GDP per Capita		-0.186	-0.120		-0.0265
		(0.290)	(0.288)		(0.290)
Total Procedure (AIH) Rate			0.00293***		0.00297***
			(0.000274)		(0.000276)
Effect in Quarter 3 of Municipal Election Year [Campaign]:					
$\delta_1 + \delta_3$	0.610***	0.608***	0.592***	0.762***	0.764***
	(0.0904)	(0.0904)	(0.0900)	(0.0908)	(0.0953)
Municipality FE	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes
Year Pair FE	No	No	No	Yes	Yes
Year	Yes	Yes	Yes	No	No
Year Squared	Yes	Yes	Yes	No	No
Control Mean	1.997	1.999	1.999	1.997	1.999
Observations	360,000	359,748	359,748	360,000	359,748

Notes: This table reports coefficients from regressions of the monthly tubal ligation rate on interactions between municipal election year and calendar quarter, as specified in Equation 2. The outcome is defined as the number of tubal ligation procedures per 100,000 women of reproductive age (15–49) residing in the municipality. The coefficient on δ_1 captures the effect in the first quarter of a municipal election year relative to years preceding municipal elections. Coefficients $\delta_2 - \delta_4$ represent marginal deviations from this Q1 baseline, so that the total effect in a given quarter is δ_1 plus the corresponding interaction term. We report $\delta_1 + \delta_3$, the total effect during the campaign quarter (Q3), at the bottom of the table. All columns include quarter fixed effects and municipality fixed effects. Columns 1-3 include quadratic time trends, while Columns 4–5 include year-pair fixed effects, where a year pair is defined as a municipal election year and the year immediately preceding it. Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Heterogeneity of Electoral Cycles in Tubal Ligation Rate (Within-Year Variation)

	(1) Northeast	(2) Other Regions	(3) Bottom Tercile Income	(4) Middle Tercile Income	(5) Top Tercile Income	(6) Nonwhite Patients	(7) White Patients	(8) Nonwhite Patients in NE	(9) White Patients in NE
δ_1: Municipal Election Year	0.169	0.0353	0.183	0.0713	-0.0270	-0.0608	-0.123	-0.218	0.0518
	(0.130)	(0.0777)	(0.135)	(0.111)	(0.0998)	(0.116)	(0.0820)	(0.209)	(0.155)
δ_2: Municipal Election Year × Q2	0.186	-0.00371	0.0800	-0.170	0.240**	0.0328	0.0537	0.270	-0.167
	(0.143)	(0.0898)	(0.136)	(0.145)	(0.112)	(0.152)	(0.116)	(0.187)	(0.243)
δ_3: Municipal Election Year × Q3	1.214***	0.267**	0.760***	0.374**	0.473***	0.475***	0.282**	1.321***	0.278
	(0.211)	(0.111)	(0.192)	(0.186)	(0.136)	(0.182)	(0.133)	(0.302)	(0.261)
δ_4: Municipal Election Year × Q4	-0.309	-0.206*	-0.498***	-0.0803	-0.127	-0.238	-0.0244	-0.186	-0.172
	(0.190)	(0.111)	(0.174)	(0.172)	(0.152)	(0.156)	(0.116)	(0.248)	(0.190)
Log GDP per Capita	-0.0339	-0.160	0.532	-0.414	-0.261	-0.416	-0.196	0.143	0.401
	(0.546)	(0.350)	(0.484)	(0.571)	(0.428)	(0.449)	(0.268)	(0.710)	(0.480)
Effect in Quarter 3 of Municipal Election Year [Campaign]:									
$\delta_1 + \delta_3$	1.382***	0.303***	0.944***	0.446***	0.446***	0.414***	0.160	1.103***	0.330*
	(0.204)	(0.0963)	(0.172)	(0.171)	(0.121)	(0.155)	(0.111)	(0.279)	(0.197)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Squared	Yes	Yes	Yes	Yes	Yes	No	No	No	No
Control Mean	1.992	2.001	1.643	1.845	2.508	2.087	1.330	1.950	0.706
Observations	102,108	257,640	119,988	119,880	119,880	143,892	143,892	40,836	40,836

Notes: This table reports coefficients from regressions of the monthly tubal ligation rate on interactions between municipal election year and calendar quarter, as specified in Equation 2. The outcome is defined as the number of tubal ligation procedures per 100,000 women of reproductive age (15–49) residing in a municipality. The coefficient on δ_1 captures the effect in the first quarter of municipal election years relative to years preceding municipal elections. Coefficients $\delta_2 - \delta_4$ represent marginal deviations from this Q1 baseline, so that the total effect in a given quarter is δ_1 plus the corresponding interaction term. We report $\delta_1 + \delta_3$, the total effect during the campaign quarter (Q3), at the bottom of the table. All columns include quarter fixed effects and municipality fixed effects. Columns 1-5 include quadratic year trends, while columns 6-9 include linear year trends. In Columns 3-5, we split the sample into terciles of GDP in 2000. Columns 6-9 restrict the analysis to 2011 onward, as we do not observe patient race in the 2007 pre-election year or earlier. We use demographic counts from the 2010 census to compute race-specific procedure rates. Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Electoral Cycles in All Hospitalizations and Selected Procedures

	All	Elective				Non-Elective	
	(1) Total Procedures Rate	(2) Vasectomy Rate	(3) Hernia Rate	(4) Gallbladder Rate	(5) Cataracts Rate	(6) Appendectomy Rate	(7) Heart Attack Rate
δ_1: Municipal Election Year	2.080*	-0.0511	0.155***	0.00547	-0.410**	-0.0239	0.0133
	(1.227)	(0.0349)	(0.0486)	(0.0339)	(0.175)	(0.0282)	(0.0197)
δ_2: Municipal Election Year × Q2	-1.820	0.0353	0.0336	0.0925**	-0.229	-0.0111	0.0211
	(1.371)	(0.0407)	(0.0598)	(0.0439)	(0.219)	(0.0365)	(0.0262)
δ_3: Municipal Election Year × Q3	3.405**	0.155***	0.342***	0.179***	0.186	-0.0148	-0.0258
	(1.651)	(0.0571)	(0.0685)	(0.0490)	(0.242)	(0.0396)	(0.0263)
δ_4: Municipal Election Year × Q4	-30.29***	0.0317	-0.673***	-0.358***	-0.146	-0.121***	-0.0554**
	(1.924)	(0.0553)	(0.0702)	(0.0499)	(0.244)	(0.0395)	(0.0278)
Log GDP per Capita	-22.58**	-0.636***	0.0329	-0.287*	-1.679**	-0.218*	-0.221***
	(9.022)	(0.156)	(0.214)	(0.166)	(0.759)	(0.112)	(0.0779)
Effect in Quarter 3 of Municipal Election Year [Campaign]:							
$\delta_1 + \delta_3$	5.484***	0.104**	0.498***	0.185***	-0.224	-0.0387	-0.0125
	(1.243)	(0.0445)	(0.0555)	(0.0392)	(0.196)	(0.0288)	(0.0210)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Squared	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control Mean	660.8	0.886	4.768	3.184	2.984	2.150	1.187
Observations	359,748	359,748	359,748	359,748	359,748	359,748	359,748

Notes: This table reports coefficients from regressions of monthly inpatient procedure rates on interactions between municipal election year and calendar quarter, as specified in Equation 2. Each outcome is defined as the procedure rate per 100,000 individuals of the medically relevant population residing in the municipality. The coefficient on δ_1 captures the effect in the first quarter of municipal election years relative to years preceding municipal elections. Coefficients $\delta_2 - \delta_4$ represent marginal deviations from this Q1 baseline, so that the total effect in a given quarter is δ_1 plus the corresponding interaction term. We report $\delta_1 + \delta_3$, the total effect during the campaign quarter (Q3), at the bottom of the table. All columns include quarter fixed effects, municipality fixed effects and quadratic year trends. Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Online Appendix

Electoral Cycles in Female Sterilization

Table A1: Summary Statistics of Inpatient Procedure Rates

	Mean	Std. Dev.	Min.	Max.
Procedure Rates for Residents of Municipality				
Tubal Ligations	2.18	10.94	0.0	916.3
Tubal Ligations (Including during C-Section)	3.26	14.03	0.0	916.3
All Procedures (AIH)	660.20	253.03	0.0	8980.9
Vasectomies	1.07	6.97	0.0	716.6
Hernia Repair	4.65	8.12	0.0	340.7
Cataracts	3.05	35.14	0.0	6191.8
Gallbladder Removal	3.39	6.86	0.0	245.6
Appendectomy	2.15	4.81	0.0	186.4
Heart Attacks	1.24	3.38	0.0	187.7
Overall Procedure Rates				
Tubal Ligations	3.72	21.09	0.0	2021.9
Tubal Ligations (Including during C-Section)	5.81	24.99	0.0	2021.9
All Procedures (AIH)	594.07	283.06	0.0	30662.7
Vasectomies	1.47	11.33	0.0	1097.2
Hernia Repair	7.65	12.31	0.0	753.0
Cataracts	9.17	55.40	0.0	10220.4
Gallbladder Removal	7.61	12.85	0.0	632.5
Appendectomy	4.35	8.81	0.0	542.2
Heart Attacks	2.30	6.31	0.0	271.1

Notes: This table reports summary statistics for monthly inpatient procedure rates at the municipality level. All rates are expressed as the number of procedures per 100,000 residents for whom the procedure is medically relevant: females aged 15–49 for tubal ligations; males aged 15–49 for vasectomies; individuals aged 55 and older for cataracts; and all residents for the remaining procedures. The "Procedure Rates for Local Residents" panel includes only patients who reside in the municipality where they received care and is limited to municipalities with hospital infrastructure. The "Overall Procedure Rates" panel includes all procedures performed in each municipality, regardless of patient residence. Observations are at the municipality-month level.

Table A2: Electoral Cycles in Tubal Ligation Rates (Year-Level Variation) — Robustness to Restriction to Municipal Election Year and Preceding Year

	(1) Tubal Ligation Rate	(2) Tubal Ligation Rate	(3) Tubal Ligation Rate	(4) Tubal Ligation Rate
Municipal Election Year	0.162***	0.168***	0.160***	0.175***
	(0.0500)	(0.0499)	(0.0500)	(0.0502)
Log GDP per Capita		0.604***	-0.186	-0.119
		(0.122)	(0.290)	(0.288)
Total Procedure (AIH) Rate				0.00295***
				(0.000272)
Municipality FE	No	No	Yes	Yes
Year	Yes	Yes	Yes	Yes
Year Squared	Yes	Yes	Yes	Yes
Control Mean	1.997	1.999	1.999	1.999
Observations	360,000	359,748	359,748	359,748

Notes: This table reports estimates of the relationship between municipal election years and tubal ligation rates, as specified in Equation 1. Observations are at the municipality-month level. The outcome is the number of tubal ligations per 100,000 women of reproductive age (15–49) residing in the municipality. The sample is restricted to procedures performed in a patient's municipality of residence and to municipalities with at least one hospital. Unlike our primary specifications, analysis in this table is restricted to observations during municipal election years and the years preceding them. Standard errors clustered at the municipality level are shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Electoral Cycles in Tubal Ligation Rates (Year-Level Variation) — Robustness to Inclusion of All Procedures and Municipalities

	(1) Tubal Ligation Rate	(2) Tubal Ligation Rate	(3) Tubal Ligation Rate	(4) Tubal Ligation Rate	(5) Tubal Ligation Rate
Municipal Election Year	0.154**	0.163**	0.191***	0.280***	0.320***
	(0.0648)	(0.0663)	(0.0661)	(0.0656)	(0.0671)
Log GDP per Capita		1.240***	-0.841**	-0.833**	-0.674*
		(0.143)	(0.353)	(0.353)	(0.345)
National Election Year				0.286***	0.209***
				(0.0659)	(0.0670)
Total Procedure (AIH) Rate					0.00727***
					(0.00136)
Municipality FE	No	No	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Year Squared	Yes	Yes	Yes	Yes	Yes
Control Mean	5.848	6.211	6.211	6.211	6.211
Observations	1,453,848	1,386,996	1,386,996	1,386,996	1,386,996

Notes: This table reports estimates of the relationship between municipal election years and tubal ligation rates, as specified in Equation 1. Observations are at the municipality-month level. The outcome is the number of tubal ligations per 100,000 women of reproductive age (15–49) residing in the municipality. Analysis in this table includes all procedures in all municipalities; compared to our primary specifications, it also includes procedures performed outside of a patient's municipality of residence, and those tubal ligations performed during a cesarean delivery. To ensure a balanced panel, the sample excludes the 1% of Brazilian municipalities created after 1998. Standard errors clustered at the municipality level are shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Electoral Cycles in Tubal Ligations (Year-Level Variation) — Results in Levels

	(1) Num. of Tubal Ligations	(2) Num. of Tubal Ligations	(3) Num. of Tubal Ligations	(4) Num. of Tubal Ligations	(5) Num. of Tubal Ligations
Municipal Election Year	0.0343***	0.0244*	0.0400***	0.0444***	0.0605***
	(0.0117)	(0.0127)	(0.0114)	(0.0106)	(0.0138)
Log GDP per Capita		0.746***	-0.213**	-0.212**	-0.115*
		(0.195)	(0.104)	(0.104)	(0.0647)
National Election Year				0.0141	-0.00583
				(0.00888)	(0.0123)
Total Procedures (AIH)					0.00823*
					(0.00436)
Municipality FE	No	No	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes
Year Squared	Yes	Yes	Yes	Yes	Yes
Control Mean	0.622	0.661	0.661	0.661	0.661
Observations	792,000	755,484	755,484	755,484	755,484

Notes: This table reports estimates of the relationship between municipal election years and tubal ligations (in levels not rates), as specified in Equation 1. Observations are at the municipality-month level. The dependent variable is defined as the number of tubal ligations performed on residents of a municipality in a given month. The sample is restricted to procedures performed in a patient's municipality of residence and to municipalities with at least one hospital. Standard errors clustered at the municipality level are shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Electoral Cycles in Tubal Ligations (Year-Level Variation) — First vs. Second Half of Sample Years

	1998 - 2008					2009 - 2019				
	(1) Tubal Ligation Rate	(2) Tubal Ligation Rate	(3) Tubal Ligation Rate	(4) Tubal Ligation Rate	(5) Tubal Ligation Rate	(6) Tubal Ligation Rate	(7) Tubal Ligation Rate	(8) Tubal Ligation Rate	(9) Tubal Ligation Rate	(10) Tubal Ligation Rate
Municipal Election Year	0.222***	0.208***	0.217***	0.248***	0.247***	0.149**	0.149**	0.149**	0.204***	0.252***
	(0.0639)	(0.0626)	(0.0632)	(0.0599)	(0.0599)	(0.0688)	(0.0693)	(0.0692)	(0.0692)	(0.0696)
Log GDP per Capita		0.402***	-0.148	-0.135	-0.137		1.093***	0.193	0.189	0.210
		(0.120)	(0.289)	(0.291)	(0.291)		(0.167)	(0.335)	(0.335)	(0.332)
National Election Year				0.120**	0.0978*				0.157***	0.111**
				(0.0546)	(0.0543)				(0.0530)	(0.0528)
Total Procedure (AIH) Rate					0.00227***					0.00480***
					(0.000247)					(0.000412)
Municipality FE	No	No	Yes	Yes	Yes	No	No	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Squared	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control Mean	1.228	1.400	1.400	1.400	1.400	2.945	2.947	2.947	2.947	2.947
Observations	396,000	359,760	359,760	359,760	359,760	396,000	395,724	395,724	395,724	395,724

Notes: This table reports estimates of the relationship between municipal election years and tubal ligation rates, as specified in Equation 1. This table splits the sample into two periods: Columns 1–5 cover 1998–2008 and Columns 6–10 cover 2009–2019. Observations are at the municipality-month level. The outcome is defined as the number of tubal ligation procedures per 100,000 women of reproductive age (15–49) residing in a municipality. The sample is restricted to procedures performed in a patient's municipality of residence and to municipalities with at least one hospital. To ensure a balanced panel, the sample excludes the 1% of Brazilian municipalities created after 1998. Standard errors clustered at the municipality level are shown in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Electoral Cycles in Tubal Ligation Rates — Robustness to Inclusion of Post-Election Year and All Years

	Inc. Post-Election Year			All Years		
	(1) Tubal Ligation Rate	(2) Tubal Ligation Rate	(3) Tubal Ligation Rate	(4) Tubal Ligation Rate	(5) Tubal Ligation Rate	(6) Tubal Ligation Rate
δ_1: Municipal Election Year	0.301***	0.303***	0.274***	0.220***	0.261***	0.234***
	(0.0571)	(0.0571)	(0.0565)	(0.0546)	(0.0552)	(0.0550)
δ_2: Municipal Election Year × Q2	-0.0614	-0.0621	-0.0279	-0.0814	-0.107	-0.0607
	(0.0678)	(0.0678)	(0.0677)	(0.0646)	(0.0654)	(0.0651)
δ_3: Municipal Election Year × Q3	0.336***	0.336***	0.365***	0.279***	0.236***	0.289***
	(0.0912)	(0.0913)	(0.0912)	(0.0847)	(0.0853)	(0.0853)
δ_4: Municipal Election Year × Q4	-0.442***	-0.442***	-0.313***	-0.457***	-0.489***	-0.328***
	(0.0883)	(0.0884)	(0.0873)	(0.0789)	(0.0802)	(0.0793)
Log GDP per Capita		-0.174	-0.113		-0.268	-0.194
		(0.249)	(0.247)		(0.269)	(0.263)
Total Procedure (AIH) Rate			0.00295***			0.00344***
			(0.000253)			(0.000314)
Effect in Quarter 3 of Municipal Election Year [Campaign]:						
$\delta_1 + \delta_3$	0.637***	0.639***	0.639***	0.499***	0.497***	0.523***
	(0.0838)	(0.0841)	(0.0840)	(0.0803)	(0.0806)	(0.0807)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Year Squared	Yes	Yes	Yes	Yes	Yes	Yes
Control Mean	2.035	2.036	2.036	2.137	2.270	2.270
Observations	540,000	539,628	539,628	792,000	755,484	755,484

Notes: This table reports coefficients from regressions of the monthly tubal ligation rate on interactions between municipal election year and calendar quarter, as specified in Equation 2. The outcome is defined as the number of tubal ligation procedures per 100,000 women of reproductive age (15–49) residing in a municipality. Unlike our primary specifications, Columns 1–3 include observations from the year following municipal elections, while Columns 4–6 include all years. The coefficient on δ_1 captures the effect in the first quarter of a municipal election year relative to nonelection years (those preceding and following municipal elections). Coefficients $\delta_2 - \delta_4$ represent marginal deviations from this Q1 baseline, so that the total effect in a given quarter is δ_1 plus the corresponding interaction term. We report $\delta_1 + \delta_3$, the total effect during the campaign quarter (Q3), at the bottom of the table. All columns include quarter fixed effects, municipality fixed effects and quadratic year trends. Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Electoral Cycles in Tubal Ligation Rates (Within-Year Variation) — Robustness to Inclusion of All Procedures and Municipalities

	(1) Tubal Ligation Rate	(2) Tubal Ligation Rate	(3) Tubal Ligation Rate	(4) Tubal Ligation Rate	(5) Tubal Ligation Rate
δ_1: Municipal Election Year	0.180	0.176	0.168	0.709***	0.731***
	(0.115)	(0.115)	(0.115)	(0.116)	(0.118)
δ_2: Municipal Election Year × Q2	0.0944	0.0963	0.0979	0.0944	0.0979
	(0.155)	(0.155)	(0.155)	(0.155)	(0.155)
δ_3: Municipal Election Year × Q3	0.314*	0.314*	0.278*	0.314*	0.278*
	(0.163)	(0.163)	(0.164)	(0.163)	(0.164)
δ_4: Municipal Election Year × Q4	-0.624***	-0.625***	-0.493***	-0.624***	-0.493***
	(0.175)	(0.175)	(0.177)	(0.175)	(0.177)
Log GDP per Capita		-0.376	-0.218		-0.138
		(0.359)	(0.358)		(0.357)
Total Procedure (AIH) Rate			0.00559***		0.00563***
			(0.00156)		(0.00158)
Effect in Quarter 3 of Municipal Election Year [Campaign]:					
$\delta_1 + \delta_3$	0.493***	0.490***	0.446***	1.023***	1.009***
	(0.127)	(0.127)	(0.127)	(0.127)	(0.131)
Municipality FE	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes
Year Pair FE	No	No	No	Yes	Yes
Year	Yes	Yes	Yes	No	No
Year Squared	Yes	Yes	Yes	No	No
Control Mean	5.035	5.034	5.034	5.035	5.034
Observations	660,840	660,468	660,468	660,840	660,468

Notes: This table reports coefficients from regressions of the monthly tubal ligation rate on interactions between municipal election year and calendar quarter, as specified in Equation 2. The outcome is defined as the number of tubal ligation procedures per 100,000 women of reproductive age (15–49) residing in the municipality. Analysis in this table includes all procedures in all municipalities; compared to our primary specifications, it also includes procedures performed outside of a patient's municipality of residence, and those tubal ligations performed during a cesarean delivery. The coefficient on δ_1 captures the effect in the first quarter of a municipal election year relative to years preceding municipal elections. Coefficients $\delta_2 - \delta_4$ represent marginal deviations from this Q1 baseline, so that the total effect in a given quarter is δ_1 plus the corresponding interaction term. We report $\delta_1 + \delta_3$, the total effect during the campaign quarter (Q3), at the bottom of the table. All columns include quarter fixed effects and municipality fixed effects. Columns 1-3 include quadratic time trends, while Columns 4–5 include year-pair fixed effects, where a year pair is defined as a municipal election year and the year immediately preceding it. Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Electoral Cycles in Tubal Ligations (Within-Year Variation) Procedure — Results in Levels

	(1) Num. of Tubal Ligations	(2) Num. of Tubal Ligations	(3) Num. of Tubal Ligations	(4) Num. of Tubal Ligations	(5) Num. of Tubal Ligations
δ_1: Municipal Election Year	0.0208	0.0187	0.0197	0.0576***	0.0804***
	(0.0140)	(0.0143)	(0.0140)	(0.0132)	(0.0191)
δ_2: Municipal Election Year × Q2	0.0289**	0.0286**	0.0381**	0.0289**	0.0381**
	(0.0141)	(0.0141)	(0.0157)	(0.0141)	(0.0157)
δ_3: Municipal Election Year × Q3	0.0937***	0.0937***	0.0900***	0.0937***	0.0900***
	(0.0196)	(0.0196)	(0.0196)	(0.0196)	(0.0196)
δ_4: Municipal Election Year × Q4	-0.0158	-0.0159	0.0239	-0.0158	0.0239
	(0.0213)	(0.0213)	(0.0368)	(0.0213)	(0.0368)
Log GDP per Capita		-0.212*	-0.158*		-0.131
		(0.119)	(0.0834)		(0.0825)
Total Procedures (AIH)			0.00792		0.00793
			(0.00495)		(0.00495)
Effect in Quarter 3 of Municipal Election Year [Campaign]:					
$\delta_1 + \delta_3$	0.115***	0.112***	0.110***	0.151***	0.170***
	(0.0195)	(0.0194)	(0.0199)	(0.0208)	(0.0260)
Municipality FE	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes
Year Pair FE	No	No	No	Yes	Yes
Year	Yes	Yes	Yes	No	No
Year Squared	Yes	Yes	Yes	No	No
Control Mean	0.597	0.597	0.597	0.597	0.597
Observations	360,000	359,748	359,748	360,000	359,748

Notes: This table reports coefficients from regressions of monthly tubal ligation procedures (in levels not rates) on interactions between municipal election year and calendar quarter, as specified in Equation 2. The dependent variable is defined as the number of tubal ligations performed for residents of a municipality in a given month. The coefficient on δ_1 captures the effect in the first quarter of a municipal election year relative to years preceding municipal elections. Coefficients $\delta_2 - \delta_4$ represent marginal deviations from this Q1 baseline, so that the total effect in a given quarter is δ_1 plus the corresponding interaction term. We report $\delta_1 + \delta_3$, the total effect during the campaign quarter (Q3), at the bottom of the table. All columns include quarter fixed effects and municipality fixed effects. Columns 1-3 include quadratic time trends, while Columns 4–5 include year-pair fixed effects, where a year pair is defined as a municipal election year and the year immediately preceding it. Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A9: Heterogeneity of Electoral Cycles in Tubal Ligation Rate (Within-Year Variation) — Robustness to Inclusion of All Procedures and Municipalities

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Northeast	Other Regions	Bottom Tercile Income	Middle Tercile Income	Top Tercile Income	Nonwhite Patients	White Patients	Nonwhite Patients in NE	White Patients in NE
δ_1: Municipal Election Year	0.422**	0.0633	0.612***	-0.0100	-0.0548	-0.165	-0.157	0.350	-0.0454
	(0.181)	(0.147)	(0.207)	(0.232)	(0.154)	(0.283)	(0.189)	(0.347)	(0.285)
δ_2: Municipal Election Year × Q2	0.675***	-0.182	0.268	0.0981	-0.0770	-0.297	-0.129	0.370	0.359
	(0.231)	(0.201)	(0.286)	(0.313)	(0.195)	(0.397)	(0.257)	(0.440)	(0.374)
δ_3: Municipal Election Year × Q3	1.448***	-0.231	0.524*	0.256	0.163	0.538	-0.187	1.627***	0.741*
	(0.267)	(0.204)	(0.299)	(0.326)	(0.209)	(0.392)	(0.312)	(0.495)	(0.433)
δ_4: Municipal Election Year × Q4	-0.893***	-0.496**	-1.430***	-0.315	-0.129	-0.715*	0.170	-0.753	-0.0722
	(0.288)	(0.219)	(0.321)	(0.348)	(0.227)	(0.406)	(0.267)	(0.475)	(0.376)
Log GDP per Capita	0.303	-0.452	1.841**	-0.616	-1.862***	0.537	-1.511**	2.991*	-2.831**
	(0.736)	(0.414)	(0.808)	(0.602)	(0.454)	(0.785)	(0.635)	(1.680)	(1.397)
Effect in Quarter 3 of Municipal Election Year [Campaign]:									
$\delta_1 + \delta_3$	1.870***	-0.168	1.136***	0.245	0.108	0.374	-0.344	1.976***	0.695**
	(0.219)	(0.154)	(0.237)	(0.249)	(0.161)	(0.296)	(0.259)	(0.400)	(0.335)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Squared	Yes	Yes	Yes	Yes	Yes	No	No	No	No
Control Mean	4.413	5.333	4.595	5.135	5.373	6.296	4.345	5.461	1.991
Observations	214,428	446,040	220,308	220,080	220,080	264,180	264,180	85,764	85,764

Notes: This table reports coefficients from regressions of the monthly tubal ligation rate on interactions between municipal election year and calendar quarter, as specified in Equation 2. The outcome is defined as the number of tubal ligation procedures per 100,000 women of reproductive age (15–49) residing in the municipality. Analysis in this table includes all procedures in all municipalities; compared to our primary specifications, it also includes procedures performed outside of a patient's municipality of residence, and those tubal ligations performed during a cesarean delivery. The coefficient on δ_1 captures the effect in the first quarter of a municipal election year relative to years preceding municipal elections. Coefficients $\delta_2 - \delta_4$ represent marginal deviations from this Q1 baseline, so that the total effect in a given quarter is δ_1 plus the corresponding interaction term. We report $\delta_1 + \delta_3$, the total effect during the campaign quarter (Q3), at the bottom of the table. In Columns 3-5, we split the sample into terciles of GDP in 2000. Columns 6-9 restrict the analysis to 2011 onward, as we do not observe patient race in the 2007 pre-election year or earlier. We use demographic counts from the 2010 census to compute race-specific procedure rates. Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$..

Table A10: Heterogeneity of Electoral Cycles in Tubal Ligations (Within-Year Variation) — Results in Levels

	(1) Northeast	(2) Other Regions	(3) Bottom Tercile Income	(4) Middle Tercile Income	(5) Top Tercile Income	(6) Nonwhite Patients	(7) White Patients	(8) Nonwhite Patients in NE	(9) White Patients in NE
δ_1: Municipal Election Year	0.0503*	0.00675	0.0160	0.0392**	0.00525	0.0107	-0.0197*	0.0552*	0.00426
	(0.0274)	(0.0165)	(0.0152)	(0.0197)	(0.0343)	(0.0170)	(0.0109)	(0.0333)	(0.00915)
δ_2: Municipal Election Year × Q2	0.0342	0.0265	0.0303	-0.00247	0.0581**	0.0112	0.0265**	-0.00411	-0.00627
	(0.0283)	(0.0162)	(0.0215)	(0.0218)	(0.0292)	(0.0126)	(0.0110)	(0.0254)	(0.00782)
δ_3: Municipal Election Year × Q3	0.215***	0.0458**	0.114***	0.0897***	0.0775*	0.0561***	0.0363***	0.109***	0.0196*
	(0.0408)	(0.0221)	(0.0275)	(0.0321)	(0.0411)	(0.0179)	(0.0134)	(0.0371)	(0.0111)
δ_4: Municipal Election Year × Q4	-0.0171	-0.0154	-0.0402**	-0.0232	0.0157	-0.0159	0.0207*	-0.0756*	-0.00549
	(0.0360)	(0.0262)	(0.0204)	(0.0260)	(0.0549)	(0.0208)	(0.0116)	(0.0414)	(0.00799)
Log GDP per Capita	-0.131	-0.214	0.0246	-0.0170	-0.290	0.107	0.0406	0.224*	0.00377
	(0.150)	(0.138)	(0.0579)	(0.0865)	(0.279)	(0.0724)	(0.0448)	(0.130)	(0.0238)
Effect in Quarter 3 of Municipal Election Year [Campaign]:									
$\delta_1 + \delta_3$	0.265***	0.0525***	0.130***	0.129***	0.0828**	0.0668***	0.0166	0.165***	0.0239**
	(0.0482)	(0.0193)	(0.0301)	(0.0345)	(0.0362)	(0.0194)	(0.0125)	(0.0406)	(0.0103)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Squared	Yes	Yes	Yes	Yes	Yes	No	No	No	No
Control Mean	0.462	0.650	0.198	0.322	1.271	0.310	0.209	0.299	0.0468
Observations	102,108	257,640	119,988	119,880	119,880	143,892	143,892	40,836	40,836

Notes: This table reports coefficients from regressions of monthly tubal ligation procedures (in levels not rates) on interactions between municipal election year and calendar quarter, as specified in Equation 2. The dependent variable is defined as the number of tubal ligations performed for residents of a municipality in a given month. The coefficient on δ_1 captures the effect in the first quarter of a municipal election year relative to years preceding municipal elections. Coefficients $\delta_2 - \delta_4$ represent marginal deviations from this Q1 baseline, so that the total effect in a given quarter is δ_1 plus the corresponding interaction term. We report $\delta_1 + \delta_3$, the total effect during the campaign quarter (Q3), at the bottom of the table. In Columns 3-5, we split the sample into terciles of GDP in 2000. Columns 6-9 restrict the analysis to 2011 onward, as we do not observe patient race in the 2007 pre-election year or earlier. We use demographic counts from the 2010 census to compute race-specific procedure rates. Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Electoral Cycles in All Hospitalizations and Selected Procedures — Robustness to Inclusion of All Procedures and Municipalities

	All	Elective				Non-Elective	
	(1) Total Procedures Rate	(2) Vasectomy Rate	(3) Hernia Rate	(4) Gallbladder Rate	(5) Cataracts Rate	(6) Appendectomy Rate	(7) Heart Attack Rate
δ_1: Municipal Election Year	1.439	0.0470	0.151**	-0.0301	-1.209***	0.0220	0.0506*
	(1.419)	(0.0453)	(0.0613)	(0.0583)	(0.217)	(0.0428)	(0.0300)
δ_2: Municipal Election Year × Q2	-0.285	-0.110*	0.142*	0.313***	-0.287	-0.0592	0.0274
	(1.038)	(0.0587)	(0.0787)	(0.0756)	(0.260)	(0.0559)	(0.0415)
δ_3: Municipal Election Year × Q3	6.414***	-0.0528	0.621***	0.274***	0.903***	0.0667	-0.0795*
	(1.370)	(0.0659)	(0.0846)	(0.0812)	(0.308)	(0.0588)	(0.0416)
δ_4: Municipal Election Year × Q4	-23.58***	-0.133*	-0.874***	-0.695***	-0.492*	-0.124**	-0.0746*
	(1.623)	(0.0704)	(0.0845)	(0.0818)	(0.293)	(0.0580)	(0.0421)
Log GDP per Capita	-28.24***	-0.463***	-0.130	-0.310**	-1.234	-0.207**	-0.121**
	(5.809)	(0.119)	(0.144)	(0.132)	(0.910)	(0.0865)	(0.0607)
Effect in Quarter 3 of Municipal Election Year [Campaign]:							
$\delta_1 + \delta_3$	7.853***	-0.00583	0.772***	0.244***	-0.306	0.0886**	-0.0289
	(1.058)	(0.0496)	(0.0649)	(0.0620)	(0.244)	(0.0416)	(0.0304)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Squared	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control Mean	591.9	1.207	7.626	6.984	8.704	4.089	2.136
Observations	660,468	660,468	660,468	660,468	660,468	660,468	660,468

Notes: This table reports coefficients from regressions of monthly procedure rates on interactions between municipal election year and calendar quarter, as specified in Equation 2. Analysis in this table includes all procedures in all municipalities; compared to our primary specifications, it also includes procedures performed outside of a patient's municipality of residence. Each outcome is defined as the procedure rate per 100,000 individuals of the medically relevant population residing in a municipality. The coefficient on δ_1 captures the effect in the first quarter of a municipal election year relative to years preceding municipal elections. Coefficients $\delta_2 - \delta_4$ represent marginal deviations from this Q1 baseline, so that the total effect in a given quarter is δ_1 plus the corresponding interaction term. We report $\delta_1 + \delta_3$, the total effect during the campaign quarter (Q3), at the bottom of the table. All columns include quarter fixed effects, municipality fixed effects and quadratic year trends. Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Electoral Cycles in All Hospitalizations and Selected Procedures — Results in Levels

	All	Elective				Non-Elective	
	(1) Total Procedures	(2) Vasectomy	(3) Hernia	(4) Gallbladder	(5) Cataracts	(6) Appendectomy	(7) Heart Attack
δ_1: Municipal Election Year	-0.133	0.00630	0.0586***	-0.0168	-0.115***	-0.0459***	0.00637
	(0.601)	(0.0159)	(0.0185)	(0.0207)	(0.0396)	(0.0161)	(0.00890)
δ_2: Municipal Election Year × Q2	-1.187**	-0.00433	0.0449**	0.0325	-0.0781*	0.0260*	0.00218
	(0.497)	(0.0176)	(0.0227)	(0.0219)	(0.0417)	(0.0150)	(0.0117)
δ_3: Municipal Election Year × Q3	0.460	0.0219	0.139***	0.0795***	0.0366	0.0279	-0.0219*
	(0.684)	(0.0224)	(0.0307)	(0.0270)	(0.0287)	(0.0177)	(0.0132)
δ_4: Municipal Election Year × Q4	-5.017***	-0.0191	-0.0980***	-0.0998***	0.0395	0.0242	-0.0216
	(0.695)	(0.0227)	(0.0341)	(0.0307)	(0.0373)	(0.0199)	(0.0137)
Log GDP per Capita	-6.884	-0.377**	0.206*	-0.604***	-0.0550	-0.355***	-0.306**
	(6.615)	(0.163)	(0.122)	(0.177)	(0.178)	(0.137)	(0.119)
Effect in Quarter 3 of Municipal Election Year [Campaign]:							
$\delta_1 + \delta_3$	0.327	0.0282*	0.198***	0.0628***	-0.0781**	-0.0180	-0.0155
	(0.643)	(0.0153)	(0.0258)	(0.0213)	(0.0376)	(0.0116)	(0.00982)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Squared	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control Mean	284.3	0.382	2.785	2.622	0.615	1.668	0.986
Observations	359,748	359,748	359,748	359,748	359,748	359,748	359,748

Notes: This table reports coefficients from regressions of monthly procedure counts (not rates) on interactions between municipal election year and calendar quarter, as specified in Equation 2. Each dependent variable is defined as the number of procedures performed for residents of the municipality in a given month. The coefficient on δ_1 captures the effect in the first quarter of a municipal election year relative to years preceding municipal elections. Coefficients $\delta_2 - \delta_4$ represent marginal deviations from this Q1 baseline, so that the total effect in a given quarter is δ_1 plus the corresponding interaction term. We report $\delta_1 + \delta_3$, the total effect during the campaign quarter (Q3), at the bottom of the table. All columns include quarter fixed effects, municipality fixed effects and quadratic year trends. Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Electoral Cycles in Tubal Ligations — First vs. Second Half of Sample Years

	1998 - 2008					2009 - 2019				
	(1) Tubal Ligation Rate	(2) Tubal Ligation Rate	(3) Tubal Ligation Rate	(4) Tubal Ligation Rate	(5) Tubal Ligation Rate	(6) Tubal Ligation Rate	(7) Tubal Ligation Rate	(8) Tubal Ligation Rate	(9) Tubal Ligation Rate	(10) Tubal Ligation Rate
δ_1: Municipal Elecion Year	0.0850	0.0828	0.0640	0.403***	0.433***	0.0978	0.0984	0.116	-0.0396	-0.00609
	(0.0777)	(0.0777)	(0.0773)	(0.0781)	(0.0867)	(0.124)	(0.124)	(0.124)	(0.120)	(0.117)
δ_2: Municipal Elecion Year × Q2	-0.00352	-0.00352	0.0112	-0.00352	0.0111	0.133	0.131	0.112	0.133	0.112
	(0.0985)	(0.0986)	(0.0984)	(0.0985)	(0.0984)	(0.139)	(0.139)	(0.139)	(0.139)	(0.139)
δ_3: Municipal Elecion Year × Q3	0.451***	0.452***	0.442***	0.451***	0.442***	0.663***	0.662***	0.651***	0.663***	0.651***
	(0.121)	(0.121)	(0.121)	(0.121)	(0.121)	(0.166)	(0.166)	(0.166)	(0.166)	(0.166)
δ_4: Municipal Elecion Year × Q4	-0.105	-0.105	-0.0358	-0.105	-0.0359	-0.429***	-0.430***	-0.294*	-0.429***	-0.294*
	(0.121)	(0.121)	(0.121)	(0.121)	(0.121)	(0.160)	(0.161)	(0.158)	(0.160)	(0.158)
Log GDP per Capita		-0.236	-0.237		-0.221		-0.290	-0.240		-0.238
		(0.340)	(0.341)		(0.344)		(0.606)	(0.600)		(0.602)
Total Procedure (AIH) Rate			0.00244***		0.00243***			0.00411***		0.00411***
			(0.000315)		(0.000315)			(0.000379)		(0.000379)
Effect in Quarter 3 of Municipal Election Year [Campaign]:										
$\delta_1 + \delta_3$	0.536***	0.535***	0.506***	0.854***	0.875***	0.761***	0.761***	0.767***	0.623***	0.645***
	(0.113)	(0.113)	(0.113)	(0.115)	(0.120)	(0.145)	(0.145)	(0.145)	(0.142)	(0.147)
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Pair FE	No	No	No	Yes	Yes	No	No	No	Yes	Yes
Year	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	No
Year Squared	Yes	Yes	Yes	No	No	Yes	Yes	Yes	No	No
Control Mean	1.403	1.404	1.404	1.403	1.404	2.889	2.891	2.891	2.889	2.891
Observations	216,000	215,856	215,856	216,000	215,856	144,000	143,892	143,892	144,000	143,892

Notes: This table reports coefficients from regressions of the monthly tubal ligation rate on interactions between municipal election year, calendar quarter, and a post-2010 indicator. The outcome is defined as the number of tubal ligation procedures per 100,000 women of reproductive age (15–49) residing in a municipality. This table splits the sample into two periods: Columns 1–5 cover 1998–2008 and Columns 6–10 cover 2009–2019. The coefficient on δ_1 captures the effect in the first quarter of a municipal election year relative to years preceding municipal elections. Coefficients $\delta_2 - \delta_4$ represent marginal deviations from this Q1 baseline, so that the total effect in a given quarter is δ_1 plus the corresponding interaction term. We report $\delta_1 + \delta_3$, the total effect during the campaign quarter (Q3), at the bottom of the table. All columns include quarter fixed effects, municipality fixed effects and quadratic year trends. To ensure a balanced panel, the sample excludes the 1% of Brazilian municipalities created after 1998. Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

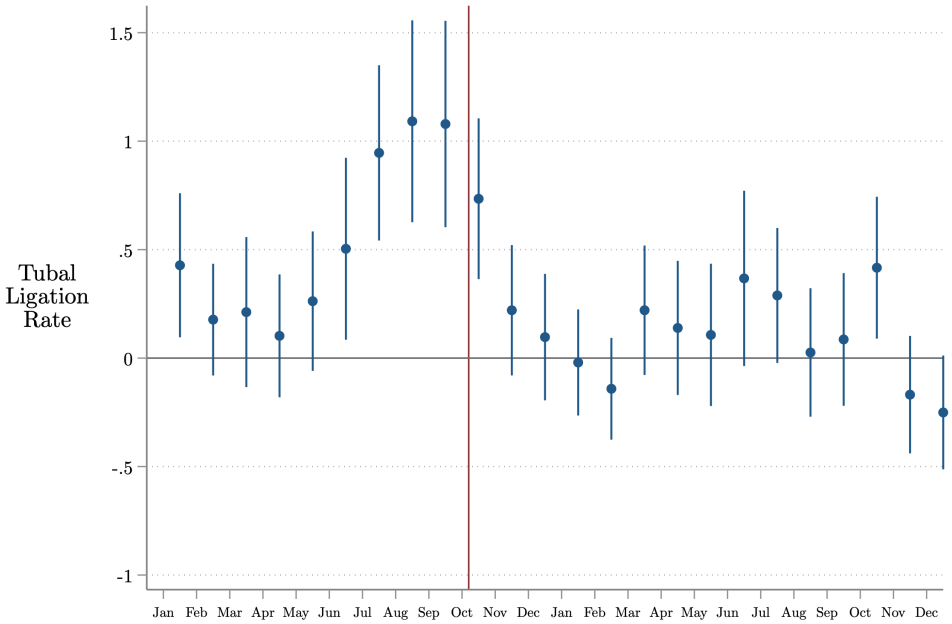
Table A14: Electoral Cycles in Tubal Ligation Rates — Quarterly Effects Over a 24-Month Window

	(1) Tubal Ligation Rate	(2) Tubal Ligation Rate	(3) Tubal Ligation Rate
Election/Post-Election Period	0.0664	0.0641	0.0575
	(0.0662)	(0.0664)	(0.0663)
× Q2 (Election Year)	0.0512	0.0502	0.0556
	(0.0760)	(0.0761)	(0.0760)
× Q3 (Election Year) [Campaign]	0.536***	0.536***	0.526***
	(0.0999)	(0.0999)	(0.0998)
× Q4 (Election Year)	-0.235**	-0.235**	-0.146
	(0.0961)	(0.0962)	(0.0955)
× Q5 (Post-Election Q1)	-0.535***	-0.542***	-0.491***
	(0.0635)	(0.0633)	(0.0622)
× Q6 (Post-Election Q2)	-0.310***	-0.317***	-0.324***
	(0.0895)	(0.0897)	(0.0897)
× Q7 (Post-Election Q3)	-0.135	-0.142	-0.170*
	(0.0916)	(0.0920)	(0.0923)
× Q8 (Post-Election Q4)	-0.120	-0.127	-0.157
	(0.104)	(0.105)	(0.106)
Log GDP per Capita		-0.218	-0.159
		(0.251)	(0.249)
AIH Rate			0.00295***
			(0.000253)
Year	72.20***	73.43***	80.03***
	(6.568)	(6.704)	(6.859)
Year Squared	-0.0179***	-0.0182***	-0.0199***
	(0.00164)	(0.00167)	(0.00171)
Municipality FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Control Mean	2.035	2.036	2.036
Obs.	540,000	539,628	539,628

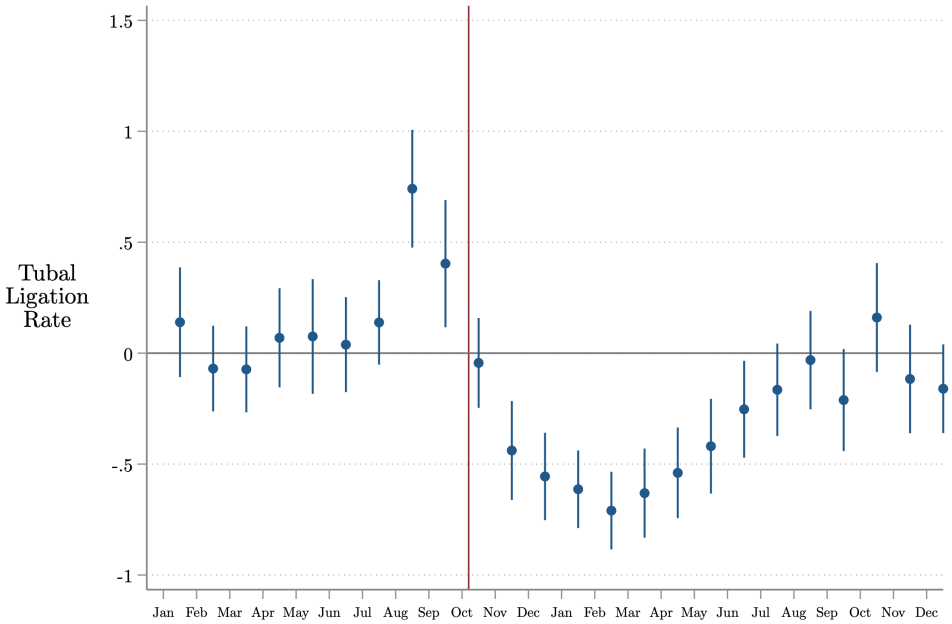
Notes: This table reports coefficients from regressions of the monthly tubal ligation rate on indicators for quarters of the municipal election year and the four quarters of the post-municipal election year. The outcome is defined as the number of tubal ligation procedures per 100,000 women of reproductive age (15–49) residing in the municipality. The coefficient on Election/Post-Election Period represents the level effect in the first quarter of the election year relative to non-election years. All subsequent coefficients (Q2–Q8) represent marginal deviations from this Q1 baseline, so that the total effect in a given quarter is the sum of the Q1 coefficient and the corresponding quarter's coefficient. All columns include municipality and quarter fixed effects, as well as a quadratic time trend. Column 2 controls for log municipal GDP per capita, and Column 3 additionally controls for the total hospitalization rate (AIH rate). Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A1: Electoral Cycle in Tubal Ligation Rates, by Mayoral Turnover

(a) Mayor Reelected



(b) Mayoral Turnover



Notes: In this figure we plot deviations from underlying trends in the tubal ligation rate. In Panel A, we estimate these deviations in municipalities where incumbent mayors are reelected, while Panel B shows deviations from municipalities that experience mayoral turnover. These deviations are estimated using a regression framework analogous to Equation 2, except with monthly rather than quarterly deviations. We include the years following municipal elections in this figure. The specification includes municipality fixed effects and quadratic time trends. 95% confidence intervals are shown, constructed using standard errors clustered at the municipality level.